
TURBO 800S/HX2.0

Operation Manual

for MC / TC / CUT / QT

Serial No. : OPE-20160607

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1. MDI Panel & External I/O Devices

This chapter explains about the panel in front of the controller and external I/O devices such as FDD and numerous ports.

- 1.1 MDI Unit**
- 1.2 Soft Key**
- 1.3 Reset Key**
- 1.4 Data Key**
- 1.5 Edit Key**
- 1.6 Cursor Key**
- 1.7 Auxiliary Key**
- 1.8 External I/O Device**



1.1 MDI unit

(1) Vertical Full Key Type



- Display: 10.4 " TFT LCD
- Resolution & Color : 640X480, 16-bit High Color

1. MDI Panel & External I/O Devices

1.2 Soft Key

This key is used to perform the screen selection and various additional functions. Among the menus on the bottom of TFT/LCD screen, black menus are used for the screen selection and gray menus are used for additional functions. Basic operation is selected using **F1** ~ **F8** keys. Returning to the main screen is made by pressing ? key left of the function keys and switching to the previous screen is made using ? key right of the function keys.

1.3 Reset Key

RESET	Reset	Resets the status and alarms related to the operation. Returns the cursor to the initial position in MDI and Edit mode. When using Full Key, use the external RESET key. In keyboard, it is the same as "ESC" key.
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Details of the operation related status reset are as follows:

- Feed instruction cancellation
- Spindle stop
- Program execution stop
- Auxiliary function M, S, T cancellation
- Tool compensation cancellation
- Target object coordinate system cancellation (Target object coordinate system 0:Maintain /1:Cancel function applied)
- Alarm reset

1.4 Data Input Key

Basic character and number keys for inputting the data such as edit, setting and parameters. In Short key type, basically, the number on the bottom of the key is entered and when the key is pressed while pressing the Shift key, the character on the upper part of the key is entered.



1.5 Edit Key

Ins	Insert	Changes the character input mode between Insert mode and Overwrite mode at the current cursor position.
Del	Delete	Deletes a character right after the current cursor position. It plays the same role as 'Delete' key in keyboard.

1. MDI Panel & External I/O Devices

Enter	Input	Used for line break in Edit window. Also, used to apply the input data to the system during setting and parameter input.
Help	G Code Help	Convenient for writing a program that it calls for help on G code entered during edition. When using a PC keyboard, F12 key can be used alternately.
Cancel	Cancel	It is not used during editing and performs the cancellation of various functions such as the warning message cancellation and data input cancellation. When using a PC keyboard, Alt key can be used alternately.

1.6 Cursor Key

↑, ↓, ←, →, and ⏏ direction keys are used to move the cursor up, down, left, or right in order to point the data input position for MDI, edition, setting and parameters. **PgUp** and **PgDn** keys are used to locate the desired address in the setting and parameter screen. It is also used to switch from the initial screen.

1.7 Auxiliary Key

Space	Space	Moves the cursor to make a blank space between words during program coding.
	Backspace	Moves the cursor one previous step and deletes the character on that position during program coding.
Home	Start of Block	Moves the cursor to the beginning of the block during program coding.
End	End of Block	Moves the cursor to the end of the block during program coding.
Ctrl	Control	When using a PC-based program, it performs the same role as Ctrl key of the keyboard.

Alt	Alternate	Performs the same role as Cancel key. When using a PC-based program, it performs the same role as Alt key in PC.
Tab	TAB	When using a Windows application program, it performs the same role as Tab key in PC.

1.8 External I/O Device

1.8.1. FDD (3.5")

Located at the bottom of MDI panel. It is used to input/output the program file using 3.5" floppy diskette. Spring cap is provided to prevent the intrusion of dust and other matters.

1.8.2 Printer Port (LPT)

Located at the bottom of MDI panel. It is used to connect the printer.

1.8.3 DNC Port (COM1)

Located at the bottom of MDI panel. It is used to transmit/receive the DNC program using RS232C.

1.8.4 Keyboard

Keyboard can be connected to KEYB'D port at the bottom of MDI panel. All functions corresponding to MDI panel keys can be used using this keyboard. It is also used to run Windows NT O/S or other Windows-based applications.

1.8.5 Mouse

Mouse is connected to the MOUSE port at the bottom of MDI panel. It is

1. MDI Panel & External I/O Devices

used to run Windows NT O/S or other Windows-based applications.



Memo



2. Manual Operation

It is the manual machine manipulation function for various settings prior to or during the automatic operation.

2.1 Machine Zero Return

2.2 JOG Feed

2.3 MPG (Handle) Feed

2.4 Manual Absolute ON/OFF

2.1 Machine Zero Return

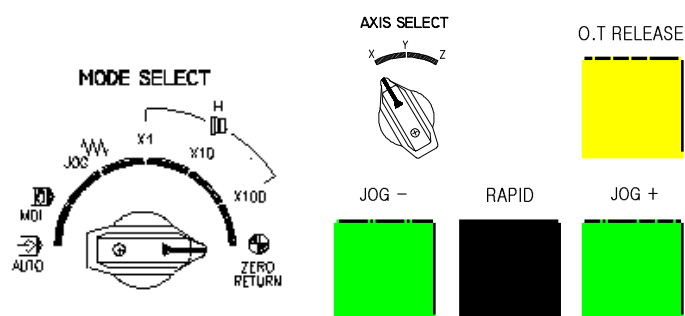
Generally, the Machine Zero Return is performed after powering on the CNC device. It is used to notify the CNC device of the origin of the machine coordinate system and represent the absolute position of the machine coordinate system. Machine Zero Return is processed by switching the operation mode to ZRN(REF1.), selecting a desired axis using the axis selection switch, and pressing the Zero Return Direction JOG Feed button.

If you press the JOG Feed button in the opposite direction of Zero Return, JOG Feeding can be made. It is useful to move the axis to where the Zero Return can be made if the axis is located near the origin.

Operation Method

1. Turn the power ON.
2. Turn the Mode Selection Switch to ZRN (ZERO RETURN, REF1).
3. After selecting an axis to perform the Zero Return, press the JOG button of the Machine Zero Return direction. Generally, for MC, perform the Zero Return on Z axis and do the same for the rest of axis in order to avoid the collision between the tool and the target object.
4. Depending on the equipment, Automatic Zero Return on all axis is supported using Automatic Zero Return button.
5. Upon completion of the Zero Return, turn the Mode Selection switch to a desired working mode.

Zero Return Related Controls



2.2 JOG Feed

It is the function for moving the machine for each axis using JOG button. Feed speed can be changed using the Manual Feed Speed Instruction switch (Manual Override function). The examples of the feed speed for each switch position are as shown in the following table. The feed speed defined for each switch position can be changed from Machining 1 item of Parameter screen. When feeding in fast feed speed, RAPID button can be used. The fast feed speed is defined for each axis and can be changed in Machining 2 item of Parameter screen.

Switch Position	Feed Speed		Switch Position	Feed Speed	
	Metric (mm/min)	Inch (inch/min)		Metric (mm/min)	Inch (inch/min)
0	0	0	8	50	2.0
1	2.0	0.08	9	79	3.0
2	3.2	0.12	10	126	5.0
3	5.0	0.2	11	200	8.0
4	7.9	0.3	12	320	12
5	12.6	0.5	13	500	20
6	20	0.8	14	790	30
7	32	1.2	15	1260	50

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
			Manual Function Setting						
PM 1160	0.0	mm/min	Manual Feeding Speed Table (#1)						
PM 1161	2.0	mm/min	Manual Feeding Speed Table (#2)						
PM 1162	3.2	mm/min	Manual Feeding Speed Table (#3)						
PM 1163	5.0	mm/min	Manual Feeding Speed Table (#4)						
PM 1164	7.9	mm/min	Manual Feeding Speed Table (#5)						
PM 1165	12.6	mm/min	Manual Feeding Speed Table (#6)						
PM 1166	20.0	mm/min	Manual Feeding Speed Table (#7)						
PM 1167	32.0	mm/min	Manual Feeding Speed Table (#8)						
PM 1168	50.0	mm/min	Manual Feeding Speed Table (#9)						
PM 1169	79.0	mm/min	Manual Feeding Speed Table (#10)						
PM 1170	126.0	mm/min	Manual Feeding Speed Table (#11)						
PM 1171	200.0	mm/min	Manual Feeding Speed Table (#12)						
PM 1172	320.0	mm/min	Manual Feeding Speed Table (#13)						
PM 1173	500.0	mm/min	Manual Feeding Speed Table (#14)						
PM 1174	790.0	mm/min	Manual Feeding Speed Table (#15)						
PM 1175	1260.0	mm/min	Manual Feeding Speed Table (#16)						
PM 1176	0.0	mm/min	Manual Feeding Speed Table (#17)						

<Parameter – Machining 1 : Manual Feed Speed Table >

PM 2759	2000.0	mm/min,deg/min	Rapid Positioning Speed Setting (1 Axis)
PM 2760	2000.0	mm/min,deg/min	Rapid Positioning Speed Setting (2 Axis)
PM 2761	2000.0	mm/min,deg/min	Rapid Positioning Speed Setting (3 Axis)
PM 2762	0.0	mm/min,deg/min	Rapid Positioning Speed Setting (4 Axis)

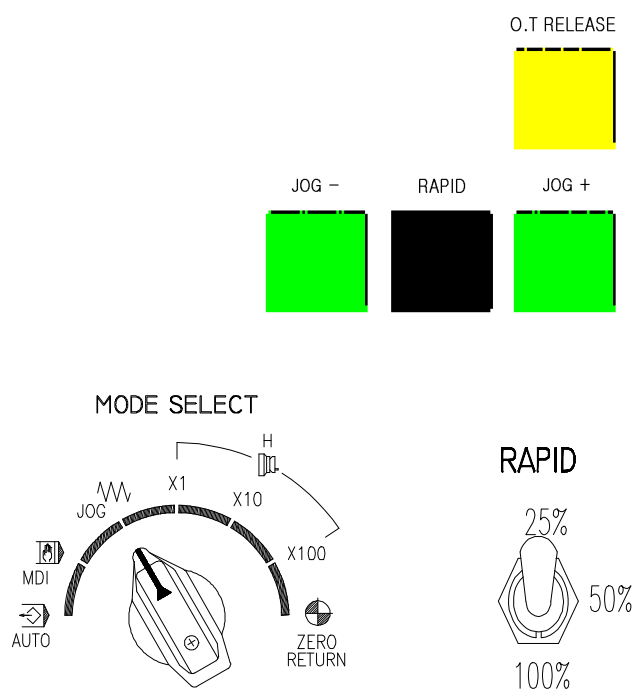
< Parameter – Machining 2 : Fast Feed Override Table >

2. Manual Operation

Operation Method

1. Turn the Mode Selection switch to JOG mode.
2. Select an axis to feed using Axis Selection switch.
3. Press the appropriate feed direction button from JOG- or JOG+ buttons.
4. When feeding in fast feed speed, press RAPID button simultaneously.
5. Since the machine moves at the maximum feed speed of each axis when using fast feeding, it is advised for the non-skilled operator to limit the RAPID Override to less than 30% in order to prevent the risk of machine collision.
6. If the machine moves out of the LIMIT switch during axis feeding (Over Travel State), Perform the Axis Feeding in the opposite direction while pressing O.T RELEASE button.

JOG Feed Related Controls



2.3 MPG (Handle) Feed

It is the function for feeding the machine for each axis as much feeding distance as wanted by rotating the MPG (Manual Pulse Generator) handle. Feed distance for one pulse can be changed and the feeding mode can be selected from X1 (basic unit feeding per 1 pulse), X10 (basic unit x 10 feeding per 1 pulse), and X100 (basic unit x 100 feeding per 1 pulse) feeding modes.

The feeding distance per 1 MPG pulse is basically the same as follows (Feeding distance per pulse can be changed by modifying 'Basic Setting Unit of MPG Feeding' parameter PM[1512~1543] for each axis from Parameter Machining 1 item).

PM 1512	0.0010	mm,deg	Basic Setting Unit for MPG Feeding (1 Axis)
PM 1513	0.0010	mm,deg	Basic Setting Unit for MPG Feeding (2 Axis)
PM 1514	0.0010	mm,deg	Basic Setting Unit for MPG Feeding (3 Axis)

< Parameter – Machining 1 : Basic Setting Unit of MPG Feeding per Axis >

Ex) Feed distance per pulse if the basic setting unit is 0.001mm.

X 1 : 0.001 [mm/pulse]

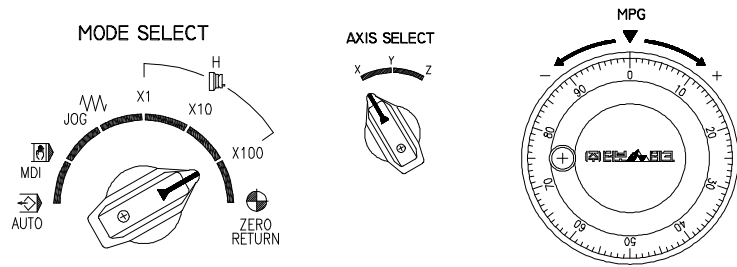
X 10 : 0.010 [mm/pulse]

X 100 : 0.100 [mm/pulse]

Operation Method

1. Turn the Mode Selection switch to MPG Mode (or Handle Mode).
2. Select the feed magnitude per pulse from MPG X1, MPG X10, and MPG X100.
3. Select an axis to feed using Axis Selection switch.
4. Feed the axis by rotating MPG toward the desired feed direction.
5. Be careful that the feeding becomes too fast if you rotate the handle too fast with the high feed magnitude set.

MPG Feed Related Controls



2.4 Manual Absolute ON/OFF

This function selects if the amount of manual operation movement is added to the coordinate values or not by the Manual Absolute switch ON/OFF of the machine controller. When the switch is at ON position, the amount of manual operation movement is added to the coordinate value and the machine is fed to the absolute coordinates instructed by the program. When the switch is at OFF position, the amount of manual operation movement is shifted to the coordinate value and the amount of manual feed is reflected to the final coordinate value.

The relationship between the amount of manual operation movement and coordinate values in relation to Manual Absolute ON/OFF is explained using an example of running the following program.

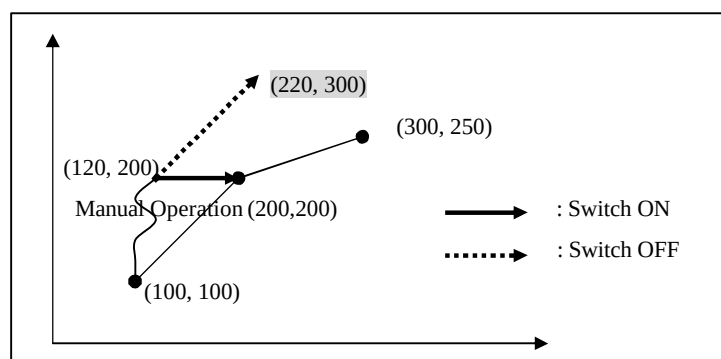
```
N1 G90 G01 X100. Y100. F100
N2 X200. Y200.
N3 X300. Y250.
```

—————▶ : Tool movement when switched ON.

.....▶ : Tool movement when switched OFF

1) Manual Operation after the Block End

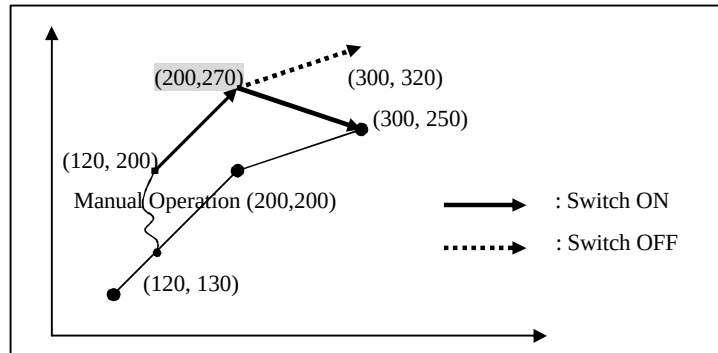
If you feed the manual operation (X-axis+20, Y-axis+100) upon completion of N1 block shift, change to AUTO Mode, and then, execute N2 block by pressing Cycle Start, the coordinate values will be as follows:



2. Manual Operation

2) Manual Operation after Feed Hold

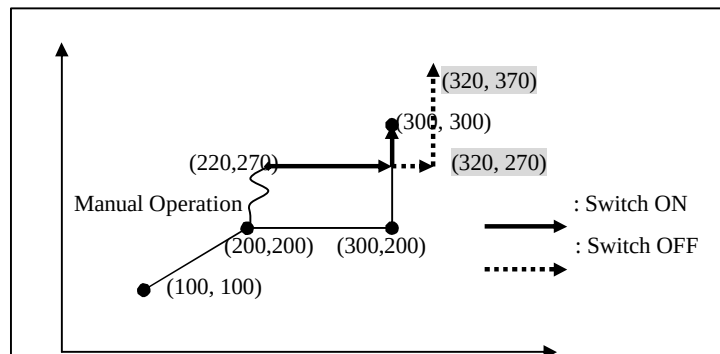
If you press Feed Hold during N1 block shift, feed the manual operation (Y-axis+70), change to AUTO Mode, and then, execute N2 block by pressing Cycle Start, the coordinate values will be as follows:



3) If the next block instruction is only for 1 axis,

If the next block instruction is made only for 1 axis, only the instructed axis is returned.

```
N1 G90 G01 X100. Y100. F100
N2 X200 Y200. : Manual feeding of X-axis+20, Y-axis+70
N3 X300.
N3 Y300.
```



4) If the next block instruction is incremental instruction,

If the next block instruction is an incremental instruction, the feeding is made as Switch OFF.



Memo



3. Automatic Operation

Machining program is used for automatic machining. There are AUTO Mode, DNC Mode, and MDI Mode operations are available. Prior to the automatic operation, the program must be carefully verified.

- 3.1 AUTO Mode Operation**
- 3.2 MDI Mode Operation**
- 3.3 Network Operation**
- 3.4 Program Restart**
- 3.5 DNC Mode Operation**
- 3.6 EDIT Mode**
- 3.7 Scheduling Operation Function**
- 3.8 Handle Interrupt**
- 3.9 Mirror Image**
- 3.10 Tool Retract/Recover**
- 3.11 Retrace Operation Function**
- 3.12 Machining Start Retract / Machining Stop
Return**



3. Automatic Operation

3.1 AUTO Mode Operation

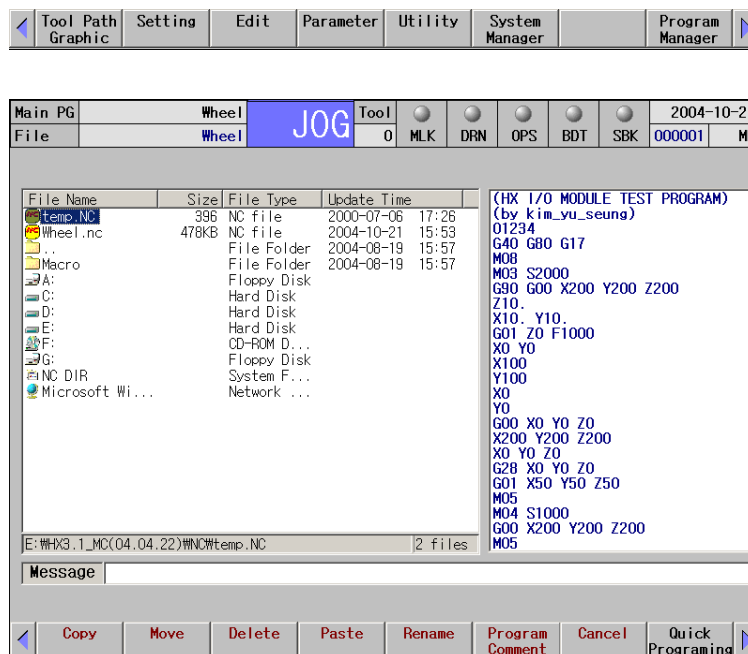
If you select the program to run (machine) from the file system where the program is registered and press Cycle Start key of the machine control panel, AUTO Mode operation is engaged and Cycle Start Lamp is turned on.

If you press Feed Hold key of the machine control panel during the AUTO Mode operation, it is temporarily halted. Pressing Cycle Start key again will resume the operation.

When RESET key is pressed, AUTO Mode operation is stopped and reset.

Operation Method

1. Turn the Mode Selection switch to AUTO Mode.
2. Press **F8** softkey to switch to **Program Manager** screen from the initial screen.



3. Select a machining program to run from the Program Management screen. (Move the cursor to the program and then, press Enter).
4. Press Cycle Start key of the control panel. AUTO Mode operation is engaged and Cycle Start Lamp is turned on. Upon completion of AUTO Mode operation, Cycle Start Lamp is turned off.
5. To stop or end the AUTO Mode operation, perform the following

procedure:

- **Stopping the AUTO Mode operation**

Press Feed Hold key of the control panel. Feed Hold Lamp is turned ON and Cycle Start Lamp is turned OFF. The machine state becomes as follows:

- ① Feeding is decelerated and finally stopped if the machine is in feeding operation.
- ② During Dwelling, it is stopped.
- ③ M, S, and T operations are stopped after the completion.
- ④ If you press Cycle Start key of the control panel while Feed Hold Lamp is ON, the machine operation is resumed.

- **Ending the AUTO Mode operation**

Pressing RESET key will stop and reset the AUTO Mode operation. If Reset key is pressed during feeding, it is decelerated and stopped.

Running AUTO Mode Operation

When the AUTO Mode operation starts, NC program is executed in the following order:

1. The instruction of 1 block is loaded from the selected program.
2. The instruction on the loaded block is parsed.
3. Execution of instruction is initiated.
4. Next block instruction is loaded.
5. It is parsed and prepared for execution. This step is so called buffering.
6. Upon completion of one block, the execution of the buffered block can be initiated.
7. AUTO Mode operation is engaged by repeating the #4, 5, and 6 steps.

Stop and Exit of AUTO Mode Operation

AUTO Mode operation can be stopped through the stop instruction and by pressing the appropriate key from the control panel.

- Stopping instructions include M00(Program Stop), M01(Optional Stop) and M02, M30 (Program End).
- Feed Hold key or **RESET** key can be used to stop the AUTO Mode

3. Automatic Operation

operation.

Program Stop (M00)

After executing the block that M00 is instructed, AUTO Mode operation is stopped. The spindle rotation is also stopped by this instruction. All modal information until the Single Block stop state is conserved. Pressing Cycle Start button will resume the AUTO Mode operation as well as the spindle rotation.

Optional Stop (M01)

AUTO Mode operation is stopped after executing the block that M01 is instructed just as in M00. However, it is applicable only when 'Optional Stop' switch of the control panel is ON.

Program End (M02, M30)

If M02 or M30 that are the final instruction on the Main Program is read, AUTO Mode operation is ended and reset.

For M30 and M02, since it is returned to the beginning of the program after all programs are completed, the same machining process can be repeated immediately after changing the target object.

If the main program is ended without M02 or M30 instruction, Alarm 82016 (Ending without M02 or M30) is generated.

Feed Hold

You can press Feed Hold key of the control panel during the AUTO Mode operation, to temporarily stop the AUTO Mode operation.

RESET

If Reset is instructed by the RESET key of the control panel/MDI panel or by the external RESET signal, AUTO Mode operation is stopped and reset. If Reset is instructed during feeding, it is decelerated and reset.

Option Block Skip (= Block Delete)

If Optional Block Skip key of the control panel is ON, all blocks containing ‘/’(slash) are ignored.

It is used to select if the current block is to be executed or not by the external signal by instructing ‘/’ at the beginning of the block. ‘/’ instruction must be placed at the beginning of the block, and otherwise, Alarm 82017 (It can be instructed only at the block beginning) is generated. If ‘/’ instruction is used at the beginning of the block and this option switch is ON, this block is skipped and proceeded to the next block.

Although Optional Block Skip can be processed by a single signal using ‘/’ instruction, it can also perform multiple steps for several signals. In this system, up to 10 external signals can be processed and those instructions are expressed as ‘/0’ ~ ‘/10’. ‘/0’ is processed by the same signal as ‘/’.

AUTO Mode Operation Related Controls



3.2 MDI Mode Operation

Programs can be created and executed in the same way as the conventional programs in MDI screen in MDI Mode.

It is used for running a simple test operation or to set the machining conditions such as spindle rotation instruction, coordinate system selection and feed instruction for resuming the program.

MDI screen is displayed by selecting the initial screen (? , F9) in MDI Mode.

Main PG	Wheel	MDI	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									000000 MM

Edit mode		Absolute		Machine	
M3 S1000		X	0.048	X	0.048
		Y	0.005	Y	0.005
		Z	3.600	Z	3.600
				Distant To Go	
				X	0.000
				Y	0.000
				Z	0.000

FEED	0.0	OVERRIDE	50
SPINDLE	0	OVERRIDE	50

Time	14:46:33	Oper. Time	0:25:19
Cyc. Time	0:00:03	T. Cyc. Time	0:00:03
DWELL	0		

Line 000001	
-------------	--

Setting	Parameter	Utility	System Manager	Clear View	Program Manager
---------	-----------	---------	----------------	------------	-----------------

Operation Method

1. Turn the Mode Selection switch to MDI Mode.
2. Write a program in the same way as for the conventional programs. One or more block instructions can be made.
3. Place the cursor to a desired block to run the program. You can start from an arbitrary block. Press Cycle Start button of the control panel to initiate the program.
4. Instruct Program End (M02, M30). The program you created is automatically reset. However, you do not have to instruct it in contrast to AUTO Mode operation.
5. If M99 is instructed, it is returned to the beginning of the program and executed repeatedly.
6. To stop or end the MDI Mode operation, perform the following

procedure:

- **Stopping of MDI Mode Operation**

Press Feed Hold key of the control panel. Feed Hold Lamp is turned ON and Cycle Start Lamp is turned OFF. The machine state becomes as follows:

- ① Feeding is decelerated and finally stopped if the machine is in feeding operation.
- ② During Dwelling, it is stopped.
- ③ M, S, and T operations are stopped after the completion.
- ④ If you press Cycle Start key of the control panel while Feed Hold Lamp is ON, the machine operation is resumed.

- **Ending the MDI Mode operation**

Pressing RESET key of MDI Panel will stop and reset the AUTO Mode operation. If Reset key is pressed during feeding, it is decelerated and stopped.

7. For the execution and stopping of MDI Mode operation, the same description of AUTO Mode operation previously mentioned in AUTO Mode Operation is applied.

Program Deletion

The program created in MDI Mode is deleted in the following cases:

- If the user parameter PA[3] 'Delete MDI Buffer during RESET' is set to 0, when Reset is instructed, program is deleted. Also, after the execution of MDI program, MDI buffer program is deleted.
- If PA[3] parameter is set to 1, MDI program is maintained until the power is OFF.
- Program is deleted by F7 Clear Screen function key.

3. Automatic Operation

Resume

If the program is resumed after some modification during stopping of MDI Mode operation, program is resumed from the block where the cursor is located.

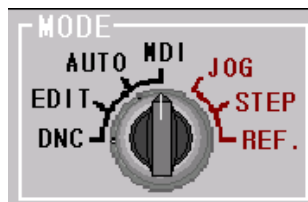
Program Modification During MDI Mode Operation

Program cannot be modified while in MDI Mode operation.

CAUTION

1. Fast Machining and reverse operation cannot be performed during MDI Mode operation.
2. Subprogram cannot be called during MDI Mode operation.
3. Program Edit window cannot be selected in MDI Mode.

MDI Related Controls



3.3 Network Operation

In this system, Network Operation function is included through LAN. Through this function, the AUTO Mode operation can be engaged from selecting the program in other PCs connected to the network.

Operation Method

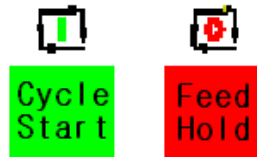
1. Press **F8** softkey to switch to Program Manager screen.
2. Select a network drive using the program selection function. (Refer to 'Chapter 7. Program Management')
3. Select a PC connected to the network.
4. Select a NC program existing in other PC.
5. Select AUTO Mode as to perform the AUTO Mode operation on general program and press Cycle Start button.

CAUTION

1. If the network is in severe state, the communication speed is slowed down as well as the AUTO Mode operation.



Network Operation Related Controls
(Same as AUTO Mode Operation)



3.4 PROGRAM RESTART

If the machining is stopped during AUTO Mode operation by the Reset instruction, power blockage, or tool failure, machining must be resumed from where it has stopped. Program Restart is used for such cases.

Operation Method

1. When resuming the operation from the block where it has stopped, press Cycle Start button after making necessary instructions such as various offset setting, spindle, and feed in MDI Mode.
2. To arbitrarily set the starting block, change to EDIT mode and locate a desired block in EDIT window. You can use Search or Shortcut function in 'F7 UTILITY' of Edit menu to easily locate the block.
3. Select MDI Mode to adjust the target object coordinate system and machining conditions, and perform the Target Object Coordinate System Selection (G54 ~ G59), Feed, Spindle Instruction, and G Code Modal Instruction.
4. Select AUTO Mode, check if the cursor is on the block to be started, and then, press Cycle Start button.
5. To start the program from the beginning, press Reset in EDIT Mode.

Block Position Selection Parameter For RESET during Program Machining

This parameter determines the cursor position that marks the program position if the machining process is stopped by Reset or Emergency Stop during AUTO Mode operation. ('PI[133] Process Block Selection during Reset' parameter among program items of Parameter Screen). You can select from maintain current block (0), move to the first block of the main program (1), or Move to the calling block of the main program (2).

PI [133]	Description
0 : Maintain	Cursor is positioned to the current block if the machining is reset. However, it is only applicable when 'Target Object Coordinate System Maintain/Cancel' parameter is set to 'Maintain' state. Enter the spindle and machining feed in MDI mode, perform Cycle Start. Change to AUTO Mode and

	perform Cycle Start again to start the machining. In this case, you do not need to find the starting position.
1 : Initial Block of the Main Program	Cursor is placed back to the starting block if the machining is Reset. Machining is started all over from the beginning. To start from the block where stopped, change to EDIT mode, locate the block or word and position the cursor, change back to AUTO mode, and then, perform Cycle Start.
2 : Called Block of the Main Program	Cursor is placed on the call block of the main program if the subprogram is Reset while called and executed. If you perform Cycle Start here, the operation is resumed from the subprogram calling.

CAUTION

1. If you are resuming the operation from an arbitrary block after the RESET, make sure that the machining conditions such as the target object coordinate system setting, spindle operation status, and feed instruction, are correct.

AUTO Mode Related Controls



3.5 DNC Mode Operation

By performing Cycle Start in DNC Mode, the program in the Host can be processed through the remote buffer for DNC communication (RS232) (DNC Mode operation). The machining program by DNC Mode operation is not saved to the system.

메인PG	0111	DNC	공구	0	MLK	DRN	OPS	BDT	SBK	2001- 9-21
진행PG	0111		0	000000	NM					

현재위치		기계위치		상대위치	
X	-100.000	X	-100.000	X	-100.000
Y	-98.042	Y	-98.042	Y	-98.042
Z	-100.000	Z	-100.000	Z	-100.000

남은거리		핸들 개입량	
X	0.000	X	0.000
Y	0.000	Y	0.000
Z	0.000	Z	0.000

현재시각	10:00:47	가공시간	0:00:04
FEED	0.0	D	0
SPINDLE	0	H	0

600	697	690	6-1	694	621	640
6-1	622	680	651	667	649	654
664	617	615	669	698	6113	611.3

공구경로	설	정	편	검	파라미터	유틸리티	시	스	탐	리	프로그램
------	---	---	---	---	------	------	---	---	---	---	------

Operation Method

1. Check the DNC communication related parameter (PP[2], PI[1 ~ 10] among user parameters). (It must coincide with Host parameter)
2. Turn the Mode Selection switch to DNC Mode.
3. Press Cycle Start button.
4. DNC transmission is initiated from the Host.

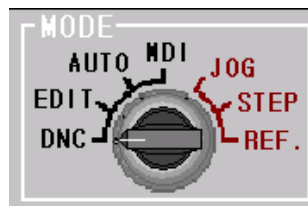
CAUTION

1. For DNC Mode operation, make sure you set DNC communication related parameters (PP[2] Com port, PI[1~2] Baud rate, PI[3~4] Parity bit, PI[5~6] Data bit, PI[7~8] Stop bit, and PI[9~10] EOT code) and the cables are correctly connected.
2. During DNC Mode operation, Program Block Display function is

disabled. You need to be careful since the verification on the machining program must be performed proactively.

3. If the subprogram call, M99 repetition function, or function that changes the program flow such as GOTO is instructed during DNC Mode operation, corresponding alarm is generated.

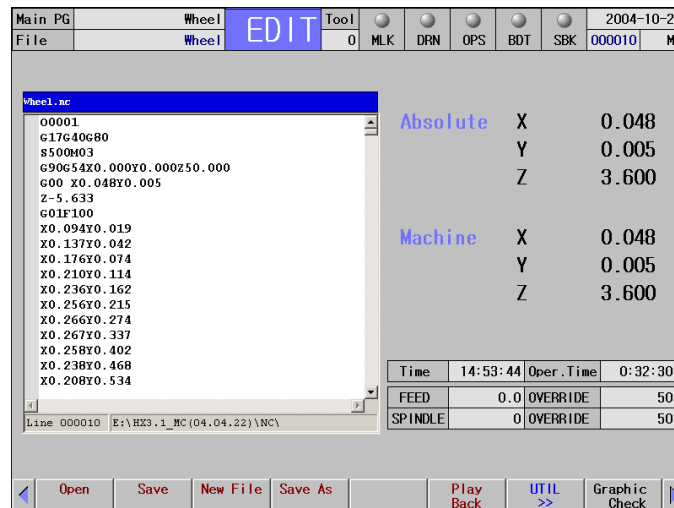
DNC Mode Operation Related Controls



3.6 EDIT Mode

EDIT mode is used to edit the selected program during AUTO Mode operation.

In this mode, you can edit the program or perform the path verification function.



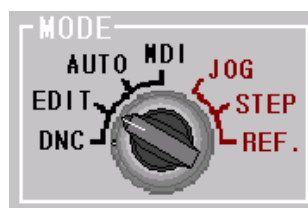
Operation Method

1. Turn the Mode Selection switch to EDIT mode.
2. Mode is changed to EDIT mode through Feed Hold during AUTO Mode operation.
3. If the Screen Change parameter (PA[323]) is set to 0, Edit window is automatically displayed.
4. If the Screen Change parameter (PA[323]) is set to 1, select F3 Edit menu from the initial menu.
5. If other program is being edited in the background, it is automatically saved and the selected program is loaded.



CAUTION

1. In EDIT mode, Background Edit or New File functions are not supported.
2. If RESET key is pressed in EDIT mode, the cursor is placed at the beginning of the program.
3. To resume the program execution from an arbitrary position, place the cursor to a desired program position in EDIT mode, change to AUTO Mode, and then, perform Cycle Start.

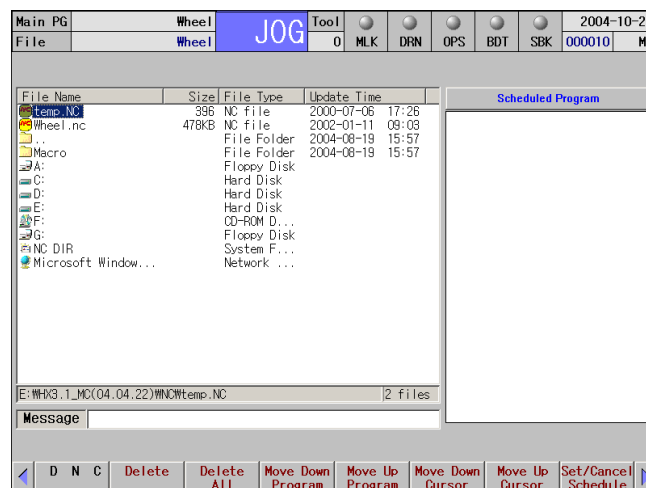
EDIT Mode Related Controls

3.7 Scheduling Operation Function

Scheduling Operation function enables the uninhabited machining for example, during night time, of the programs sequentially selected and registered.

Scheduling Operation Method

1. Switch to Scheduling Operation screen.(? Initial Screen → F5 Utility → F6 Scheduling)
2. Select and register all programs for scheduled machining.
3. Determine the execution order of the registered programs.
4. Set the Scheduling Operation Mode by pressing “Set/Release Scheduling” menu.



5. Turn the Mode Selection switch to AUTO Mode.
6. Press Cycle Start button.
7. If all scheduled programs are completed successfully, the Scheduling Operation mode is released and the program is changed to the program initially selected by the system.
8. If RESET or EMG-STOP key is pressed during Scheduling Operation, Scheduling Operation Mode is maintained, but the current machining is stopped. If Cycle Start key is pressed, Scheduling Operation is resumed.
9. To release the Scheduling Operation Mode during Scheduling Operation, press RESET key and select “Set/Release Scheduling” menu in

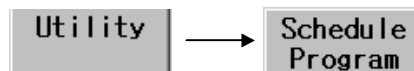
Scheduling Operation screen.

10. If all programs selected from the Scheduling Operation are executed, the Scheduling Operation Mode is automatically released.

Detailed Operation of Scheduled Machining

- **Switching to Scheduling Operation Screen**

Change the screen as shown below from the initial screen.



- **Scheduling Program Selection**

1. Using MDI key

Place the cursor on the program to select it using Up, Down, Left, or Down key.

Press Input key to register the selected program to the Scheduling Program pane on the right.

Continue to select and register the program that you wish to sequentially process.

2. Using Keyboard

Place the cursor on the program to select it using Up, Down, Left, or Down key.

Press Enter key to register the selected program to the Scheduling Program pane on the right.

Continue to select and register the program that you wish to sequentially process.

3. Using Mouse

Double click on the program to register.

Continue to double-click and register the program that you wish to sequentially process.

File Name	Size	File Type	Update Time	Scheduled Program
temp.NC	396	NC file	2000-07-06 17:26	
Wheel.nc	478KB	NC file	2004-10-21 15:53	
...		File Folder	2004-08-19 15:57	
Macro		File Folder	2004-08-19 15:57	
A:		Floppy Disk		
C:		Hard Disk		
D:		Hard Disk		

3. Automatic Operation

- **Deleting A Registered Scheduling Program**

Press F2 key. Or, click on the menu shown below at the bottom of the screen.



- **Deleting All Registered Scheduling Programs**

Press F3 key. Or, click on the menu shown below at the bottom of the screen.



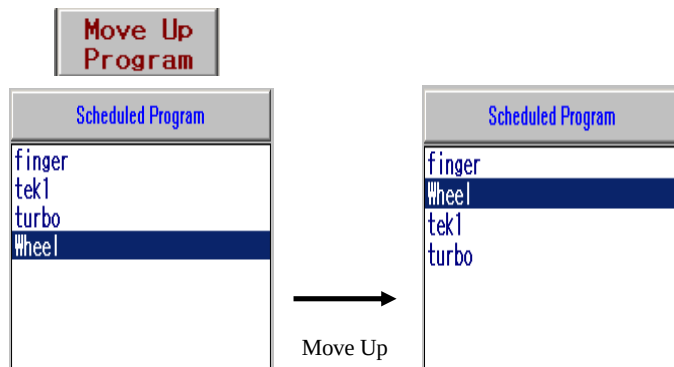
- **Bringing Down the Registered Scheduling Program**

Press F4 key. Or, click on the menu shown below at the bottom of the screen.



- **Bringing Up the Registered Scheduling Program**

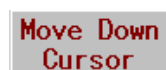
Press F5 Key. Or, click on the menu shown below at the bottom of the screen.



- **Moving Cursor in the Scheduling Program Registration Window**

Moving the cursor Down or Up

Press F6 / F7 key or click on the menu shown below at the bottom of the screen.



- **Setting or Releasing the Scheduling Operation Mode**

This function sets or releases the Scheduling Operation Mode.

When you press F8 Set/Release Scheduling key, it is set and the following warning message is displayed.

Warning Message:

F_92012[1/ 1] : Scheduling production mode.

When you press F8 key once again, it is released and the above warning message is removed.

Set/Cancel
Schedule

CAUTION for Creating Ladder to Use Scheduling Operation

The machining program selected by the system is initiated by the external Cycle Start signal. After the completion of machining by M30 instruction, you need to start the machining of the next scheduled program using **F07.00** signal. Since **F07.00** (SCST) signal is maintained as high for 100msec, you need to have **F07.00** included in the Cycle Start when creating a Ladder (G map).

3.8 Handle Interrupt

MPG feeding can be performed by overlapping the manual feeding during the AUTO Mode operation.

Handle Interrupt Axis Selection Signal

When you turn MPG if the Handle Interrupt Axis Selection signal is ON of the axis to perform the Handle Interrupt during AUTO Mode operation, the Handle Interrupt is performed.

CAUTION

1. Although the amounts of Handle Interrupt move is determined by the MPG rotation and MPG feeding multiplier (x 1, x 10, x M, x N), it may be dangerous to apply the Handle Interrupt with high MPG Move Rate. Be careful if you are using a large multiplier.

Relationship with Each Function

The relationship between the Handle Interrupt move and various functions is as follows:

Signal	Relationship
Interlock	Interlock is valid. If Interlock is ON, machine does not move.
Mirror Image	Mirror Image is invalid. Even if Mirror Image is ON, the Interrupt can be applied toward Plus(+) direction.

Position Display

The relationship between the Handle Interrupt move and various position display data is as follows:

Display	Relationship
Current Position (Absolute)	The current position (=Absolute Coordinates) by the Handle Interrupt does not change.
Relative Coordinates (Relative)	Relative Coordinates vary by the amount of Handle Interrupt.
Machine Coordinates (Machine)	Machine Coordinates vary by the amount of Handle Interrupt.

Amount of Move Display

When you press Page Up key after selecting the initial screen, the amount of move by the Handle Interrupt is displayed.

Clearing the Amount of Move

The amount of Handle Interrupt move is automatically reset to 0 when reset, Program End (M30 or M02), or the new target object coordinate system is set (G54~G59).

3.9 Mirror Image

If the Mirror Image switch on the Machine Control Panel is ON, the mirror image can be applied to the corresponding axis move in AUTO Mode operation.

Operation Method

1. Press Single Block switch and stop the AUTO Mode operation. This step is not necessary when operating with Mirror Image from the beginning.
2. Press the corresponding axis switch among the Mirror Image switches of Machine Control Panel.
3. Turn the Mode Selection switch to AUTO Mode, and press Cycle Start button to initiate the AUTO Mode operation.

Description of Operation

For more information on the Mirror Image switch, refer to the documentation provided by the machine manufacturer. If there is no Mirror Image switch available, the Mirror Image function using G code (G51) can be used instead.

CAUTIONS

1. For the feed direction of manual operation and the drilling cycle, feeding direction cannot be reversed.
2. The Mirror Image function by the Control Panel during AUTO Mode operation is only applied to G00, G01, G02, G03, and G31.x blocks.

NOTE) The Mirror Image function by the program can be automatically applied to each axis feeding even though G51 function is not used nor G51 is not directly instructed within the program by the Modal Scale Setting function (PI[149] or PI[108~116] among the program items of Parameter).(The Mirror Image function is applied to the axis that the scale factor is a negative value).

3.10 Tool Retract / Recover

Tool Retract / Recover function is used to change the tool during machining or measure the machined object. If the Tool Retract signal (TLESC) is instructed during AUTO Mode operation, the axis feeding is stopped. Up to 10 points are automatically memorized internally to NC while retracting the axis to the required position by changing to the Manual Mode. If the Tool Recover (TLRTN) signal is instructed after changing back to AUTO Mode upon completion of tool change or object measurement, the machine sequentially returns 10 points to its original position where the Tool Retract was started and the axis feeding is stopped. CNC now awaits for the signal such as Cycle Start.

Tool Retract / Recover Related Control Signals

Signal Type	Signal
Tool Retract Signal	TLESC [G79.10]
Tool Recover Signal	TLRTN [G79.11]
Tool Retract / Recover Mode Signal	TLMODL [F64.08]
Tool Retract in Progress Signal	TLESCL [F64.09]
Tool Recover in Progress Signal	TLRTNL [F64.0A]

Tool Retract Operation Method

1. Turn the [Tool Retract] switch On during AUTO Mode operation. Current block operation is stopped as well as AUTO Mode operation. In this state, Retract & Recover function can be initiated.
2. Change to Manual operation mode and move the tool. Manual JOG feed and MPG feed are possible.
3. Perform Tool Retract on one axis after another. Two or more axis selection signal cannot be specified simultaneously.
4. As you move each axis in manual mode, the controller memorizes the axis move path up to 10 points. When you select and move a certain axis to a stop and then, do the same procedure for another axis, the axis switching position is memorized. If the path memorized position exceeds 10, subsequent axis switching position is not memorized at all.

3. Automatic Operation

Tool Recover Operation Method

1. After performing the necessary actions such as the tool change, the corresponding tool needs to be returned to the position where it is retracted.
2. To return the tool to the retracted position, change to AUTO Mode, and turn the [Tool Recover] switch of the control panel to ON. Tool is returned in a DRY RUN speed.
3. Tool is returned to the original position where the retraction started through 10 points positions memorized. (Single Block can be applied)
4. During returning, Feed Hold is valid.
5. Upon completion of tool returning, press Cycle Start button to resume the rest of the work by the program.

CAUTIONS

1. When you retract the axis in manual mode after changing the Tool Retract signal to HIGH, you need to use Tool Retract/Recover Mode signal TRMODL [F64.08] in PLC to select only one axis, not two at the same time. Axis selection switching position is automatically memorized internally to NC.
2. When you instruct RESET, the data on the corresponding position is lost and the Tool Retract Mode is also cancelled.

Tool Retract/Recover Mode Signal TLMODL [F64.08]

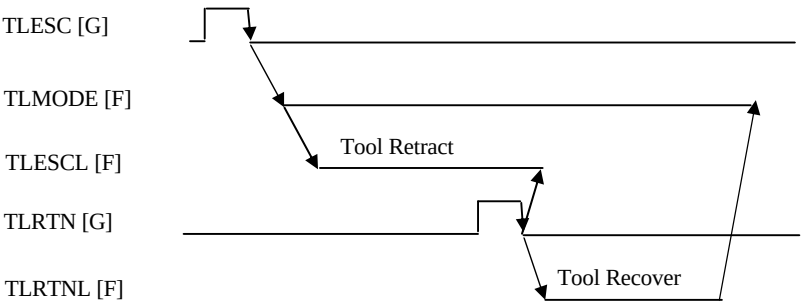
It is the confirmation signal that notifies the Tool Retract/Recover mode, and maintained to HIGH from the tool retraction to returning.

Tool Retract in Progress Signal TLESCL [F64.09]

It is the confirmation signal that notifies the Tool Retract is in progress and maintained to HIGH until TRRTN signal [G79.11] is instructed.

Tool Recover in Progress Signal TLRTNL [F64.0A]

It is the confirmation signal that notifies the Tool Recover is in progress and maintained to HIGH until TRRTN signal [G79.11] is instructed.



3.11 Retrace Function

Using this Retrace function, the reverse operation can be made by retracing the path according to the program instruction through the [Retrace Operation] switch manipulation during AUTO Mode operation and returning to the starting position of the Retrace Operation can also be made. Upon returning to the original position where the Retrace Operation is initiated, feeding is resumed according to the program instruction.

In this system, the Retrace Operation is possible up to 50 blocks.

Retrace Operation Control Signals

Signal Type	Signal
Retrace Operation ON/OFF Signal	RVS (Input) [G39.8]
Retrace Operation in Progress Signal	RVSL (Output) [F30.8]

Description of Operation

1. If the Retrace Operation signal (RVS) is turned ON during AUTO Mode operation, the machine stops after the current block processing is completed. If the Cycle Start signal is instructed, the Retrace Operation is performed as many blocks as specified by PP[51] parameter.

PP 51	50	Max IPO Hold Buffer Size
-------	----	--------------------------

2. If the Retrace Operation signal (RVS) is turned OFF during Retrace Operation, the machine stops after the current block processing is completed. If the Cycle Start signal is instructed, the forward Operation is performed.
3. The confirmation signal (RVSL) becomes HIGH while the Retrace Operation is in progress with RVS signal ON and becomes LOW while the forward operation is in progress with RVS signal OFF.

CAUTIONS

1. To perform the Retrace Operation or the forward operation immediately without completing the current block processing, instruct Feed Hold signal and set RVS signal to HIGH or LOW after checking

the Feed Hold stopping.

2. To have the operation started right after changing the RVS signal to HIGH or LOW, instruct Cycle Start signal in a few seconds after checking Feed Hold or Single Block Stop in PLC.
3. Feed Hold and Single Block functions are all valid during Retrace Operation.
4. M, S, and T Code are normally processed during the Retrace Operation. If not required, process it using RVSL signal in PLC.
5. Retrace Operation is available only for G00, G01, G02, and G03 codes.
6. Retrace Operation is only available in AUTO Mode operation, but not in MDI operation.

3.12 Machining Start Retract / Machining Stop Return

This function has been prepared to retract to the machining start point or return to the machining stop point using PLC and Custom Macro.

Machining Start Retract / Stop Return

To retract to the machining start point during AUTO Mode operation, press the Machining Start Retract switch.

After performing the necessary actions such as the tool change or replacement at this position, press the Machining Stop Return switch to automatically return to the position where the machining has stopped.

CAUTIONS

1. The first block of the AUTO Mode operation program must start with M code (It can be arbitrarily set to the macro parameter) that calls for a macro program. (Memorizes the program start point at the 4th origin).
2. If the RESET or EMG-STOP is instructed during AUTO Mode operation, Machining Stop Return cannot be performed.

Macro Program that remembers the starting point at the 4th Origin.

#8064 = #6205

#8065 = #6206

M99



Memo



4. Test Operation

This function is to check whether the made program is perfect or not without unreasonable operation of the system. Unlike the graphic function, you can verify through actual operation.

- 4.1 Machine Lock**
- 4.2 Feed Override**
- 4.3 Rapid Traverse Override**
- 4.4 Dry Run**
- 4.5 Single Block**

4.1 Machine Lock

You can perform machining without moving the machine. It is operated only by the coordinates on the screen so you can use this function in the stage of verification before actual machining work.

How to operate

1. Turn the switch of Machine Lock on
2. There is no movement of the machine for all the operations.
3. You can perform either manual operation or auto as you want.

Special notes

1. M. S. T function is performed under the status of machine lock.
2. If you perform the function of machine lock under the condition that return to the reference position is not made, auto operation is not possible (may be possible, depending on the maker of the machine)
3. Even though the indicating value of the machine location is changed by auto operation and manual operation under the condition that the switch of Machine Lock is on, the indicating value of the machine location, after the switch of Machine Lock is turned off, indicates the actual location of the machine again.
4. During auto operation, it is not cancelled by turning the switch of Machine Lock on

Relevant console box



4. Test Operation

4.2 Feed Override

This is to operate while changing the cutting feedrate by 10% from 0% to 250%. During actual cutting, you can adjust feed override suitably for the cutting status of the machining parts. Depending on the machine maker, override can be adjusted up to maximum 32 steps.

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PM 2891	0	%	Interpolation Feed Override Table (#1)						
PM 2892	10	%	Interpolation Feed Override Table (#2)						
PM 2893	20	%	Interpolation Feed Override Table (#3)						
PM 2894	30	%	Interpolation Feed Override Table (#4)						
PM 2895	40	%	Interpolation Feed Override Table (#5)						
PM 2896	50	%	Interpolation Feed Override Table (#6)						
PM 2897	60	%	Interpolation Feed Override Table (#7)						
PM 2898	70	%	Interpolation Feed Override Table (#8)						
PM 2899	80	%	Interpolation Feed Override Table (#9)						
PM 2900	90	%	Interpolation Feed Override Table (#10)						
PM 2901	100	%	Interpolation Feed Override Table (#11)						
PM 2902	110	%	Interpolation Feed Override Table (#12)						
PM 2903	120	%	Interpolation Feed Override Table (#13)						
PM 2904	130	%	Interpolation Feed Override Table (#14)						
PM 2905	140	%	Interpolation Feed Override Table (#15)						
PM 2906	150	%	Interpolation Feed Override Table (#16)						
PM 2907	0	%	Interpolation Feed Override Table (#17)						
PM 2908	0	%	Interpolation Feed Override Table (#18)						

How to operate

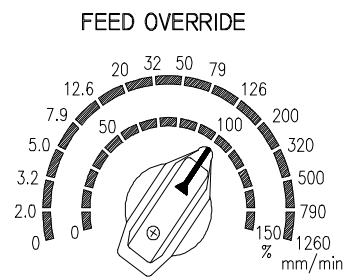
1. Rotate the switch of Feed Override before or during operation to put it on the % that you want.

Special notes

1. During screw machining, feed override gets ignored (100%) and the machine moves at the nominated feed.
2. In case the function of Override Cancel is on, feed override is fixed at 100%.



Relevant console box



4. Test Operation

4.3 Rapid Traverse Override

This is to operate while gradually changing the feedrate by 0%, 25%, 50% and 100% during rapid traverse. Depending on the machine maker, rapid traverse override can be adjusted up to maximum 32 steps. It is possible to set the rapid feedrate for each axis on the screen, “Machining 2, Parameter”.

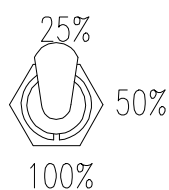
PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PM 2759	2000.0	mm/min,deg/min	Rapid Positioning Speed Setting (1 Axis)						
PM 2760	2000.0	mm/min,deg/min	Rapid Positioning Speed Setting (2 Axis)						
PM 2761	2000.0	mm/min,deg/min	Rapid Positioning Speed Setting (3 Axis)						
PM 2762	0.0	mm/min,deg/min	Rapid Positioning Speed Setting (4 Axis)						
PM 2791	100	%	Rapid Positioning Override Table (#1)						
PM 2792	10	%	Rapid Positioning Override Table (#2)						
PM 2793	50	%	Rapid Positioning Override Table (#3)						

How to operate

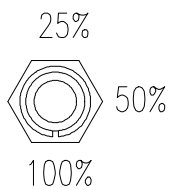
1. Put the switch of Rapid on the % that you want before/during operation.
2. Rapid traverse override is performed when:
 - ① G00 is carried out,
 - ② Positioning during cycle code in operation,
 - ③ G27 ,G28, G30 are carried out, and
 - ④ Manual rapid traverse is performed.

Relevant console box

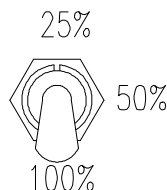
RAPID



RAPID



RAPID



4.4 Dry Run

This is used when you intend to check the moving path of the tool at the given speed under the condition that there is no work-piece. The feedrate under dry run follows manual JOG feedrate that is selected.

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PM 1160	0.0	mm/min	Manual Feeding Speed Table (#1)						
PM 1161	2.0	mm/min	Manual Feeding Speed Table (#2)						
PM 1162	3.2	mm/min	Manual Feeding Speed Table (#3)						
PM 1163	5.0	mm/min	Manual Feeding Speed Table (#4)						
PM 1164	7.9	mm/min	Manual Feeding Speed Table (#5)						
PM 1165	12.6	mm/min	Manual Feeding Speed Table (#6)						
PM 1166	20.0	mm/min	Manual Feeding Speed Table (#7)						
PM 1167	32.0	mm/min	Manual Feeding Speed Table (#8)						
PM 1168	50.0	mm/min	Manual Feeding Speed Table (#9)						
PM 1169	79.0	mm/min	Manual Feeding Speed Table (#10)						
PM 1170	126.0	mm/min	Manual Feeding Speed Table (#11)						
PM 1171	200.0	mm/min	Manual Feeding Speed Table (#12)						
PM 1172	320.0	mm/min	Manual Feeding Speed Table (#13)						
PM 1173	500.0	mm/min	Manual Feeding Speed Table (#14)						
PM 1174	790.0	mm/min	Manual Feeding Speed Table (#15)						
PM 1175	1260.0	mm/min	Manual Feeding Speed Table (#16)						
PM 1176	0.0	mm/min	Manual Feeding Speed Table (#17)						
PM 1177	0.0	mm/min	Manual Feeding Speed Table (#18)						

How to operate

1. Remove the work-piece if necessary
2. Turn the switch of Dry Run on
3. Adjust the feedrate of dry run by use of the switch of Manual Feedrate command
4. Check the moving path of the tool, while performing auto operation.

Relevant console box

4.5 Single Block

This makes it possible to perform the program by one block during auto mode operation, DNC mode operation and MDI mode operation.

While the macro program is carried out, you have to check the function of Single Block after setting the parameters - 'PI[83] Whether the macro program single block stops or not' and 'PI[84] Whether there is macro program block

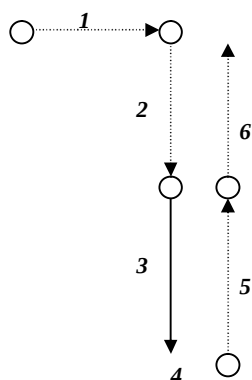
4. Test Operation

indication’.

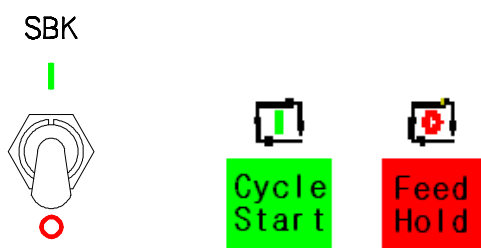
PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PA 0002	1		Macro Program Edition (0:Not Allowed, 1:Allowed)						
PI 0083	0		Macro program Stop on Single Block (0:Allowed, 1:Not Allowed)						
PI 0084	0		Macro Program Block Display (0:Not Display, 1:Display)						

How to operate

1. Turn the toggle switch of SBK (Single Block) on.
2. Press the button of Cycle Start, and the function will be performed for one block.
3. Continue to press the button of Cycle Start, and it will be performed by one block.
4. In case of cycle code, movement of one block is performed as per the following number.



Relevant console box





Memo



11. Utility

This is explanation of the convenient utility that is used in HX.

11.1 DNC

11.2 Scheduling

5.1 Emergency Stop

This is to stop the operation of the machine because of any errors with it or any dangerous situation.

How to operate

1. Press the button of Emergency Stop.
2. The message, “The emergency switch is pressed.” appears. This message is indicated through the function of handling PLC alarm message so it can be changed by a user.
3. In order to cancel it after checking the status, rotate the button of Emergency Stop to the right.

Related console box



5.2 Overtravel / Soft Limit

In case return to the reference position is not performed or the set soft limit is larger than the actual stroke of the machine, if it is out of the machine stroke, it goes into the status of overtravel.

In case return to the reference position is performed and soft limit is correctly set at the machine stroke, if it is commanded out of the stroke, it becomes the status of soft limit alarm. Under the status of soft limit, if you move the axis in the opposite direction, the alarm is cancelled immediately. It is possible to correct the soft limit setting at 'Machining 2' in 'Parameter' (PM[3378 ~ 3409] for whether to check by axis and PM[3410~3472] for the ranges of check).

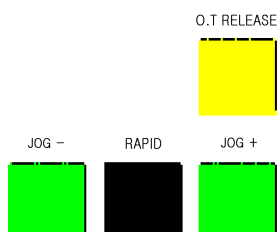
PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PM 3378	1		Soft Limit Usage Status (0:Use, 1:No Use) (1 Axis)						
PM 3379	1		Soft Limit Usage Status (0:Use, 1:No Use) (2 Axis)						
PM 3380	1		Soft Limit Usage Status (0:Use, 1:No Use) (3 Axis)						
PM 3381	0		Soft Limit Usage Status (0:Use, 1:No Use) (4 Axis)						
PM 3410	-300.00	mm,deg	Soft Limit Area #1 (1 Axis)						
PM 3411	190.00	mm,deg	Soft Limit Area #2 (1 Axis)						
PM 3412	-300.00	mm,deg	Soft Limit Area #1 (2 Axis)						
PM 3413	145.00	mm,deg	Soft Limit Area #2 (2 Axis)						
PM 3414	-180.00	mm,deg	Soft Limit Area #1 (3 Axis)						
PM 3415	190.00	mm,deg	Soft Limit Area #2 (3 Axis)						
PM 3416	0.00	mm,deg	Soft Limit Area #1 (4 Axis)						
PM 3417	0.00	mm,deg	Soft Limit Area #2 (4 Axis)						

If the machine goes into the status of overtravel out of the soft limit, an alarm happens and the operation stops.

How to operate

1. In order to cancel overtravel, change the mode to JOG or MPG with pressing the button of O.T. Release, and move the relevant axis into the stroke.
2. In order to cancel soft limit, manually move the relevant axis into the stroke.

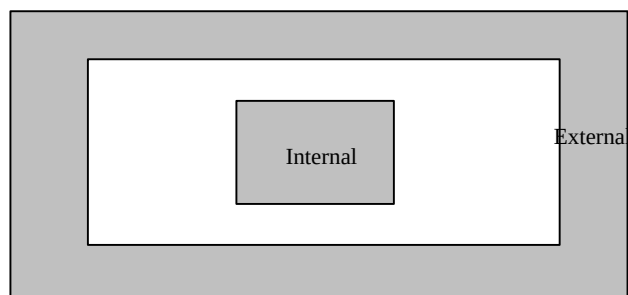
Relevant console box



5.3 Stored Stroke Check

This function is to alarm in case the tool invades the zone of stored stroke. The stored stroke is divided into the inside and the outside as shown on the below figure. Setting the stored stroke is possible at 'Machining 2, Parameter' (PM[3474] for whether to apply the stored stroke or not, PM[3475] for selection of the inside/the outside, and PM[3476~3481] for setting stored stroke). Whether to apply the stored stroke or not or change of it is possible by G22 X_ Y_ Z_ I_ J_ K_, G23.

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PM 3474	0		G22 Prohibition Area Usage Status (0:Valid, 1:Invalid)						
PM 3475	0		G22 Prohibition Area Internal/External Setting (0:Internal,1:External)						
PM 3476	0.00	mm	G22 Prohibition Area Extent of X Axis (#1)						
PM 3477	0.00	mm	G22 Prohibition Area Extent of X Axis (#2)						
PM 3478	0.00	mm	G22 Prohibition Area Extent of Y Axis (#1)						
PM 3479	0.00	mm	G22 Prohibition Area Extent of Y Axis (#2)						
PM 3480	0.00	mm	G22 Prohibition Area Extent of Z Axis (#1)						
PM 3481	0.00	mm	G22 Prohibition Area Extent of Z Axis (#2)						



How to operate

1. Put 0 or 1 into Setting the inside/the outside for stored stroke of parameter, and also input the location on the machine coordinate system for setting by axis.
2. It is possible to set stored stroke with G22 X_ Y_ Z_ I_ J_ K_.
3. If this stored stroke is invaded during operation under the condition that G22 is set, an alarm happens.
4. In order to cancel the alarm, change the mode into manual operation mode and move the invading axis out of the stored stroke.

4. Test Operation



Memo



6. Program Edit

This is a function to create NC program for machining, to correct the already made program, to delete, or to add. This system possibly works for the program available in the outside device like FDD or network.

6.1 Create and Select Edit Program

6.2 Save Edit Program and Save under New Name

6.3 Basic Use of Editor

6.4 Delete, Copy and Move Block

6.5 Find and Change

6.6 Directly Go to Line

6.7 Book Mark

6.8 Play Back

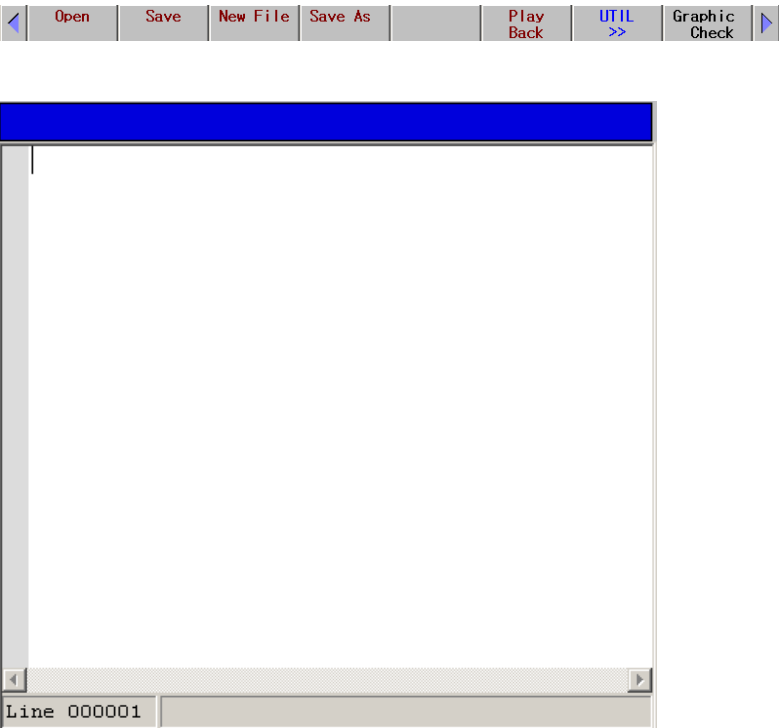


6.1 Create and Select Edit Program

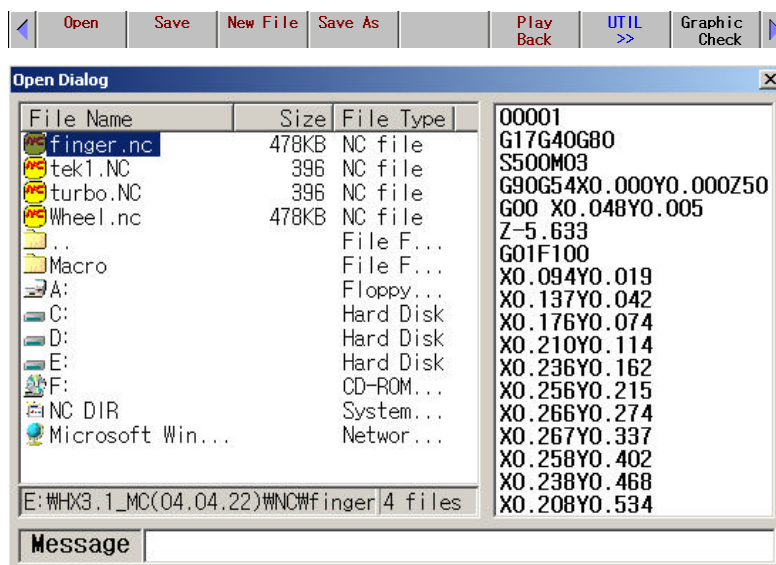
This is to create a new program or to select an already made program.

How to operate

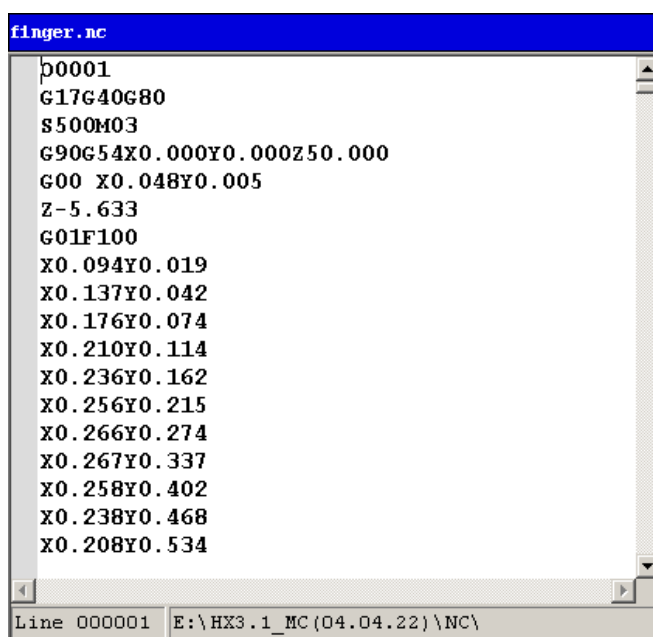
- 1. On the Initial Screen, press **F3** in order to go to the screen of **Edit**.
- 2. Press **F3**, the menu of **New File** , and an edit window for editing a new file with no file name appears as shown on the below.



- 3. Once the program is made out, press Save and input the name of the program. At this time, the extension of NC is created and it is saved in the NC folder under the system folder.
- 4. Press the menu of Open, **F1**, and the open status, which shows the file list as shown on the below figure, will appear.



5. Select a program in the list by use of the cursor or mouse, and press **ENTER** or double-click the mouse.

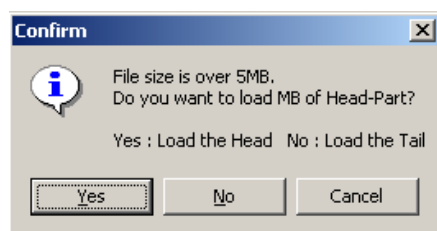


Reference 1

If you change the mode to EDIT without saving the newly made file, the contents that have been made so far are automatically saved under the file name of '_AUTOSAVE.NC', which is created in the NC folder under the system folder. And the program selected at AUTO mode is loaded on the EDIT screen.

Reference 2

1. In case the file that you intend to open is over 5MB (3MB in case of CE), the following message will appear on the screen.



2. At this time, if you press **Yes (Y)**, the former part of the file is read, while if you press, **No (N)**, the latter part of the file is read. If you don't use the mouse, use **TAB** to move to other item.

3. The following is an example that the latter part of the file is read.

A screenshot of a CNC program editor window. The title bar says "tek1.NC". The main area displays a list of G-code and M-code commands. The commands are: G00 X0 Y0 Z0, X200 Y200 Z200, X0 Y0 Z0, G28 X0 Y0 Z0, G01 X50 Y50 Z50, M05, M04 S1000, G00 X200 Y200 Z200, M05, G01 X100 Y100 Z100 M03, G00 X0 Y0 Z0, M05, M03 S3000, G00 X50, Y50, Z50, M05, M04 S1500. At the bottom, there is a status bar that reads "Line 000001 E:\HX3.1_MC\{04.04.22}\NC\".



6.2 Save Edit Program and Save Under New Name

This is to save the newly made program or the existing program after edit.
Please make sure that the edited/corrected program is not saved automatically.
If you want to save, always press **F2** , the menu of Save.

How to operate

1. Press **F2** , the menu of save, in order to save after opening the previously made file and editing. The file is saved under the previous name. You can check the saving status through the progress bar at the bottom of the editor.



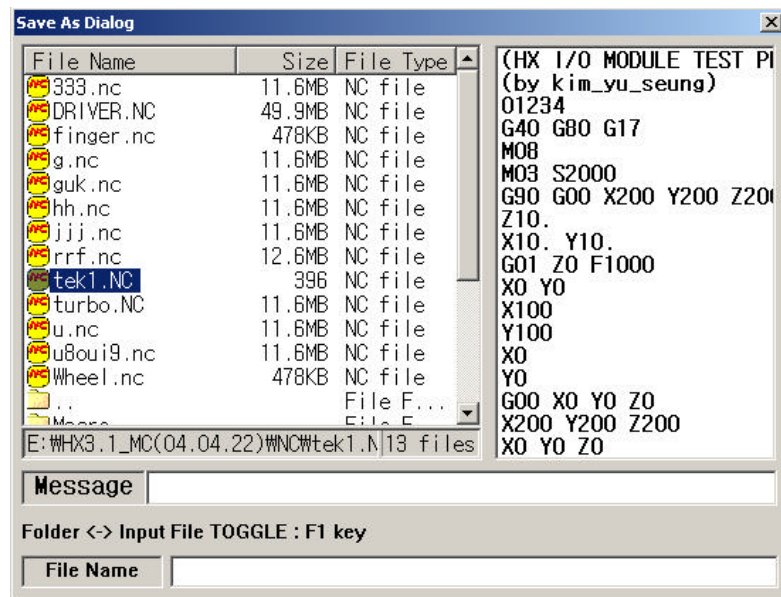
Reference 1

If you save after opening a large file (either former part or latter part) and editing, the unopened part is automatically saved too together, which may take quite long time. You can check the progress through the progress bar at the bottom of the editor.



2. If you want to save the edit program under a new name, press **F4** , the menu of Save under new file name , and the following dialog box for file saving will appear on the screen.





- The dialog box for file saving is composed of the part to select a folder and the part to input a new name. **F1** is used for moving between these two. If you select a file (previous file) in the part of selecting a folder, it is saved under the file name. If you put the file name you want for saving into the part to input the file name, it is saved under the file name (the folder in which it is saved is that appearing on the window of folder selection).

Reference 2

In case of making a new file and saving it, it is saved unconditionally in the form of saving a new file (possible to select a folder and to input a file name)

6.3 Basic Use of Editor

This is to actually edit and correct. The details are different for each function, but it is often used together in order to edit and correct.

How to operate

1. Insert letters

- ① Move the cursor to where you want to insert by use of `?` , `?` , `?` , `?`

```
G00 |X100
```

```
G00 Y20 |X100
```

- ② Input the contents to insert (Y20). The letters are input in front of the cursor and accordingly, the cursor goes behind.

2. Delete letters

- ① Locate the cursor in front of the letters that you want to delete, by use of `?` , `?` , `?` , `?`

```
G00 Y20 |X100
```

- ② Press **DEL** , and a letter after the cursor is deleted.

```
G00 Y20 |100
```

3. Replace letters

- ① Same as the case of using a general editor on Windows, if you input letters with **Ins** pressed, it becomes the Insert mode. Press **Ins** one more time to cancel the Insert mode and input letters. Then, the newly input letters get to cover the previously input letters.

6.4 Delete, Copy and Move Block

This is to delete, copy and move several blocks during editing and correcting.
It is possibly performed with the same keys for general document edit that are used on Windows.

How to operate

1. Select block/Cancel block

- ① Locate the cursor on the place that a block will start .

```
|G00 X100  
G01 Y10.1  
Z10.3
```

- ② With **SHIFT** pressed, press **?** , **?** , **?** , **?** , Page Up, or Page Down to select a block that you want. (you can select a block by dragging the mouse

```
G00 X100  
G01 |Y10.1  
Z10.3
```

- ③ In order to cancel the block, use the directional keys without pressing **SHIFT** .

2. Delete block

- ① Select a block that you will delete through the above process of 2.

```
G00 X100  
G01 |Y10.1  
Z10.3
```

- ② Once a block is selected, press **DEL** .

3. Copy block

6. Program Edit

- ① Select a block through the process of 2.

```
G00 X100
G01 |Y10.1
Z10.3
```

- ② Once a block is selected, press the key of Copy, **Ctrl** .

4. Cut block

- ① Select a block through the process of 2. 2

```
G00 X100
G01 |Y10.1
Z10.3
```

- ② Once a block is selected, press the key of cut, **Ctrl** and **X** together.

```
|Z10.3
```

5. Paste block

- ① Move the cursor to the place where you will paste a block, by use of **?** , **?** , **?** , **?**

```
G00 X100
G01 Y10.1
Z10.3
|
```

- ② Press the key of paste, **Ctrl** and **V** together.

```
G00 X100
G01 Y10.1
Z10.3
G00 X100
G01 Y10.1|
```

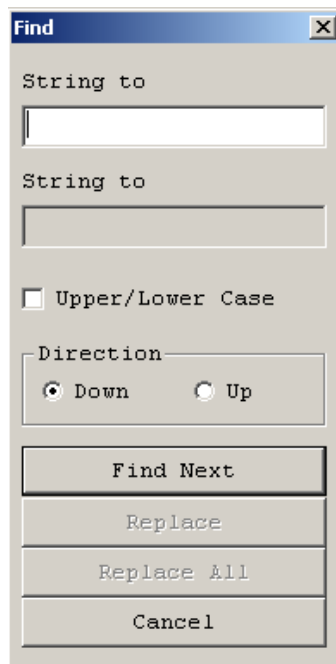
6.5 Find and Change

It is possible to find a character string in the program that you want and also to change it with other character strings. After using this function once, you can continuously find the next character string by use of **ENTER**.

When the program is divided by statement label for starting in the middle, this function is also used for finding the statement label.

How to operate 1 (Find)

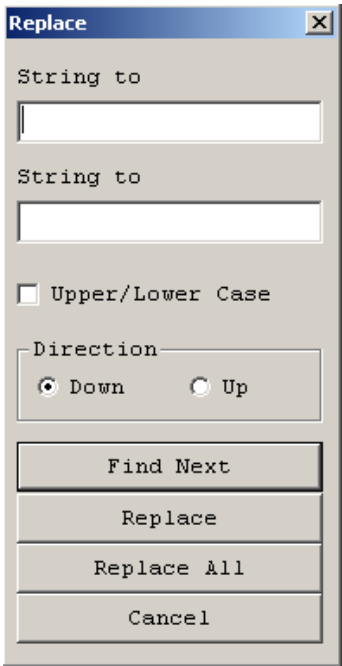
1. Press the menu of **UTIL >>**, **F7**, on the Edit screen to activate the sub-menu, and then, press the menu of **Find**, **F1**. And the dialog box will appear on the screen.



- 2. In item of ‘Character string to find’, input the character string that you want to find and then, press **ENTER** . Then, finding starts (it is okay to press the button of ‘Find next’ by use of mouse.) In order to finish the finding, press **CANCEL** or RESET.
- 3. Check (Space key) the item, ‘Character and small letter separation’, and it reads character/small letters separately and performs finding.

How to operate 2 (Change)

- 1. Press the menu of **UTIL >>** , **F7** , on the edit screen to activate the sub-menus, and then, press the menu of **Change** , **F2** . And the dialog box will appear on the screen.



2. In the item, 'Character string to find', input the character string that you want to find, and in the item, 'Character string to change', input new contents for change. Once input is completed, press **TAB** to select the button of **Change** and then, press , **ENTER** . And the function of change is performed. If you press **ENTER** without changing the focus with **TAB** , the function of **Find next** will be automatically performed. Just like the function of 'Find', the function of Find Next is used when moving to the next character string to find without performing the function of 'Change'.
3. Check (Space key) the item, 'Character and small letter separation', and it reads character/small letters separately and performs finding and changing. '
4. '**Change All**' is used for changing all in the editor at a time, not one by one. In case of a large file, when only the former or the latter of a document is opened, the function of Change All works. However, it may take quite long time. Therefore, in case of performing the function, 'Change All' of a large file, you can check the progress through the progress bar at the bottom of the editor.



Reference

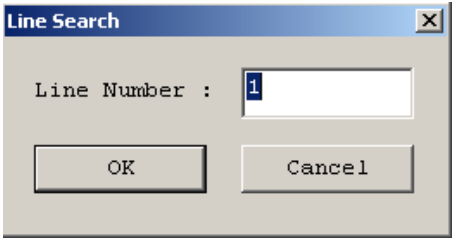
When working on the function of Find or Change, you can change the location of the dialog box appearing on the screen by pressing **Ctrl** and **?, ?, ?, ?** together.

6.6 Directly Go to Line

This is to move the cursor to the block you want.

How to operate

1. Press the menu of **UTIL >>**, **F7**, on the edit screen to activate the sub-menus, and then, press the menu of **Directly go**, **F3**. And the dialog box asking the line that you want to find will appear.



2. In the input window, input the line number you want to find and press **ENTER**. And the cursor will move to the head of the block that you want to find.

6.7 Book Mark

This function is used for designating the location that you want to read later and then, fast moving to it.

How to operate

Press the menu of **UTIL >>**, **F7**, on the edit screen to activate the sub-menus. At the place that you will use book mark, press **Ctrl** and **F4** together. And as shown on the below, a light blue circular mark will appear on the left. This is a book mark.



1. In order to cancel the book mark, go to the place where the book mark is designated and then, press **Ctrl** and **F4** together. (designation and cancellation of book mark are performed in the form of toggle).
2. Press **F4** to go to the place for next book mark.
3. In order to go to the place of the previous book mark, press **Shift** and **F4** together.

6.8 Play Back

With Play Back, you can make a program by registering the machine location of X, Y and Z with the program.

How to operate

1. Move the machine by JOG or MPG to the location you want.
2. Input the name of axis and locate the cursor behind it in order to designate the coordinate axis first.

X|

3. Press the menu of **Play Back**, and the current machine location will be input.

X0.000

4. For other axes, repeat the above processes of 2 and 3.

X0.000 Y-100.000 Z-50.000



Memo



7. Program Management

This function is to select NC program saved in memory as a processing program or to manage. Here the memory means HDD of PC. In addition, it works on the external interface like FDD and network.

7.1 Select Processing Program

7.2 Copy and Move Program

7.3 Delete Program

7.4 Change Program Name

7.5 Description of Program (Comments)

7.6 Use Network

7.1 Select Processing Program

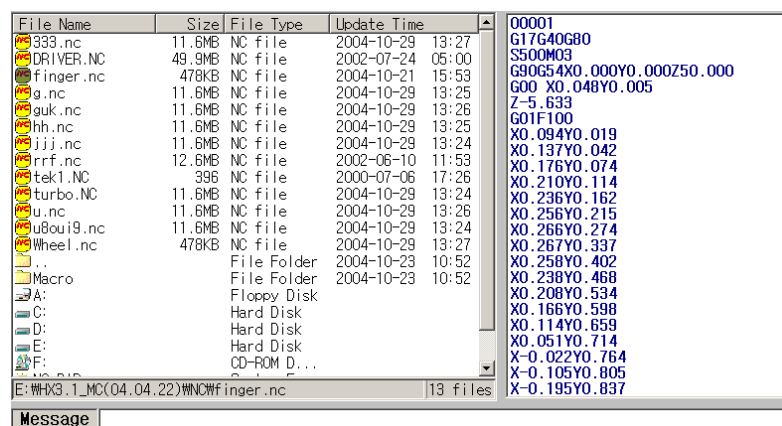
This is to select a program for processing and register it as the main program.

It is performed similarly with the function of selecting edit program in edit.

The selected main program is used for AUTO mode operation. .

How to operate

1. Press the menu of **Program Management** on the Initial Screen, **F8** .
2. The Preview window on the right shows the contents of NC program in about 20 lines. By use of this window, you can see the contents of NC program in advance.



3. Locate the selection bar on the program that you want to select by use of **↑**, **↓** . At this time, the message window at the bottom shows comments saved in each program.
4. Press **Enter** on the program that you will select (or double-click with mouse), and the Initial Screen will appear automatically and the program will be selected.
5. It is NC folder under the program folder that you get into first. In order to move to other folder, select the folder you want to move to and press **Enter** . In order to return to the previous folder, select “ “ folder. In case of moving to the initial NC folder while searching for other drive, select “NC DIR”.
6. After the program is selected and the screen is changed into Initial Screen, see the program name on the left top, which will be the newly

7. Program Management

selected program.

Main PG	Wheel
File	Wheel

7.2 Copy and Move Program

This is to copy or move the existing program in/to other folder or drive. It is possible to copy in the same folder and it is also possible to freely copy or move to the computer connected with network.

How to operate

1. Press **F8** on the Initial Screen to change to the screen of **Program Management**.
2. Locate the select bar on the program you want by use of **?**, **?**, **?**, **?**.
3. Press the menu of **Copy**, **F1**, to copy and press **F2** to move. Then, the following message will appear. .

Message **Copy KEY is selected.**

Message **Move KEY is selected.**

4. Select the folder you want to copy in. (Press the menu of **Cancel**, **F7**, to cancel the copy or the move.) Once the folder is selected, press the menu of paste, **F4**.
5. If there is a file of the same name in the selected folder, the following message will appear.

Message **This file already exists. Replace/Save As/Cancel (R/S/C)**

6. Press **R** to overwrite with the same name and press **S** to save under other name. Press **C** to cancel. In case of selecting to save under other name, the message window at the bottom of the screen becomes available for input as shown on the below and you need to input a new file name. Input the file name you want there (if you don't type the extension, "NC" is added automatically) and then press **Enter**.

Message **Art1004**

7. Program Management

7.3 Delete Program

This is to delete the existing program.

How to operated

1. Use **F8** on the Initial Screen to go to the screen of **Program Management**.
2. Locate the select bar on the program to delete by use of **?** , **?** , **?** , **?** .
It is possible to select a program available on the network.
3. Press the menu of **Delete**, **F3** .
4. The Y/N question window as shown on the below will appear. Input Y if you want to delete or N if you don't want.

Message **Delete file really? (Y/N)**

5. If you press Y, the program deleted from the program list will disappear.
6. In case of deleting the selected program for AUTO mode operation, all the contents of the program will be deleted while only the name remains. .

7.4 Change Program Name

This is to change the name of the existing program.

How to operate

1. Use **F8** on the Initial Screen to go to the screen of **Program Management**.
2. Locate the select bar on the program to change by use of **?**, **?**, **?**, **?**.
It is possible to select a program available on the network. **?**
3. Press the menu of **Change Name**, **F5**.
4. It is changed into the status that you can change the file name as shown on the below (like the shape of finder). Input a new name here and press **Enter**.

 tek1.NC	396
 turbo.NC	11.6MB
 u.nc	11.6MB

5. The existing name in the program list will be changed into the new name.

7. Program Management

7.5 Description of Program (Comment)

This is to read brief description of the program or to newly edit. Together with the screen of Preview, it is a convenient function to grasp the contents of the program in advance before selecting a program. It is possible to register detailed description of the program, not only its name.

How to operate

1. Use **F8** on the Initial Screen to go to the screen of **Program Management**.
2. Move to the program that you want by use of **?**, **?**, **?**, **?**
3. Press the menu of **Description of Program**, **F6**. And the message window as shown on the below will be activated to be available for edit. You write down description of the program here and press **Enter**. Description of the program is saved under the same file name, but with the extension of 'NCT'.

Message	Test Program
---------	--------------

7.6 Use Network

It is possible to copy a program from other PC connected on the network or to copy a program in the system into other PC. It means the basic use of network. In case of using the function of network operation, select a program here to perform AUTO mode operation.

How to operate

1. Use **F8** on the Initial Screen to go to the screen of **Program Management**.
2. Locate the select bar on [Microsoft Windows Network] by use of **↑**, **↓**, **→**, **←**.
3. Press **Enter**, and the following list will appear on the list window.
Here select the domain you want (the domain to which your computer belongs is usually selected).

File Name	Size	File Type	Update Time
WORKGROUP		Network Domain	
A:		Floppy Disk	
C:		Hard Disk	
D:		Hard Disk	
E:		Hard Disk	
F:		CD-ROM Drive	
NC DIR		System Folder	
Microsoft Wi...		Network System	

4. Press **Enter** in the domain you want, and the list of the connected PC's will appear. Here select the PC you want and perform the above process from 7.1 to 7.5.



Memo

8. Graphic Function

You can monitor the current status of machining by use of graphics and also can check the path of the tool in advance without operating the machine actually.

8.1 View Tool Path Real TPG

8.2 Test Tool Path Test TPG

8.1 View Tool Path (Real TPG)

This is to check the moving path of the tool on a real time basis by use of the graphic function. It can check the path of the tool that is machined, not only the location on a real-time basis. This function doesn't work during MDI mode operation.

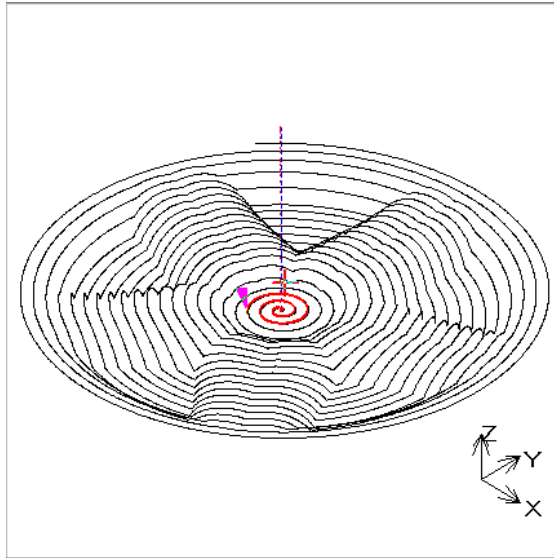
In order to normally check the graphics, you should accurately divide the parameter, 'PP[20] Type of Machine for TPG Application (0 for milling, 1 for lathe and 2 for NCT class), of HMI parameters and set. This parameter is related with the NC program interpreter.

The screen of View Tool Path shows the tool path on a real-time basis based on the work-piece coordinate system (white solid line for cutting feed, yellow dotted line for rapid traverse and green solid line for screw cutting). In case of setting the parameter of Preview (PA[409] Real TPG Screen Mode: To be indicated after Preview), the path of the tool is drawn in advance when the screen is changed into the screen of View Tool Path. You can check the path of machining, indicating the tool's location (PA[407] Indication of Chasing Real TPG Path) and trace (PA[408] Trace Indication for Chasing Real TPG) by use of cursor.

8. Graphic Function

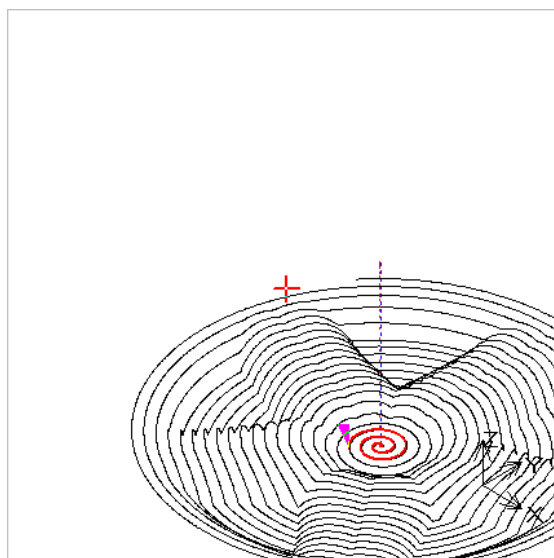
How to operate

1. Press the menu of **View Tool Path**, **F1**, on the Initial Screen.
2. If it is under AUTO mode operation or DNC mode operation, you can see the graphic window as shown on the below by using the menu, 'F1 View Tool Path' in the initial menu. It appears on the left of the screen (the parameter regarding indication will be explained in the end of this chapter).

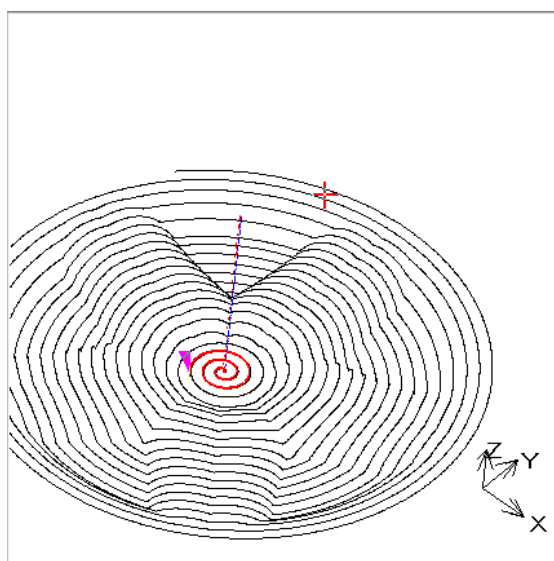


3. The followings are functions available in the menu, View Tool Path:
 - ① Increase: Press **PgUp**. (Press the left button of mouse and drag in)
 - ① Decrease: Press **PgDn**. (Press the left button of mouse and drag out)
 - ② Zoom All: Press the function key of **Zoom All**, **F5**. All the pictures will appear in the screen.
 - ③ Set View Direction: Press the menu of **Set View Direction**, **F3**. The planes set with the parameter will appear in order. For example, if the planes are set in the order of XYZ, XY, YZ, ZX, these four views appear in turn.
 - ④ Move Screen (up, down, left and right): The screen is moved as per the following process:
 - Press **Set Center for View**, **F4**.

- The mark of cross will appear in the middle of the graphic window as shown on the below:



- Set the location of the new center by use of ? ,? ,? ,? .



- After moving, press **ENTER** or the menu of **Set Center for View**, **F4** , and the set location of the cross will become the

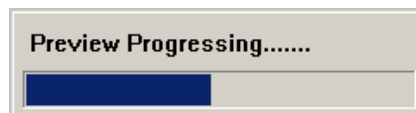
8. Graphic Function

new center of the screen (double-click the mouse on the place that you want)

- ⑤ Rotation: Rotation is possible in two directions – up and down, and left and right. Direction key $\uparrow$, $\downarrow$, $\leftarrow$, $\rightarrow$ can make it (in case of using the mouse, rotation is possible, having an arbitrary axis as the center).
- ⑥ Delete Screen: Press the menu of **Delete Screen**, **F7**, and the paths which have been indicated on the screen so far will be deleted.

4. Preview

- ① The function of Preview is to check the tool path in advance before actual machining in order to confirm whether machining path is being progressed in the desirable path.
- ② For preview, you need to set the relevant items with the parameter (the relevant parameter will be explained in the below).
- ③ The function of Preview works if it is set with the parameter in case of moving to the screen of **View Tool Path**. While the function of preview is working, the progress window as shown on the below will appear so you can check any progress in the performance of preview.



- ④ If you press the menu of, **Delete Screen**, **F7** when the function of Preview is selected so that a picture is drawn, only the part of indicating machining chase (trace indication) will get cleared.

Relevant parameter and edit parameter

1. Set View Direction

- ① It is possible to select total 10 directions of view (PA[364-373])
- ② Set the view direction that is not used at 0.
- ③ Number (1, 2, 3...) in the order of use and set the

direction of view.

- ④ For example, set that only the XZ plane is used for lathe or only the ZX plane is used for vertical lathe.
- ⑤ On the screen of View Tool Path or Check Tool Path, press the menu of Set View Direction , **F3** , and the direction of view will be changed in the order selected.
- ⑥ Initial View Direction is the parameter that is used when there is no set view direction. A value from 0 to 9 should be input.

2. Parameter related with screen indication

- ① It is possible to select total 10 view directions (PA[364-373])
- ② If the item, ' Whether there is an indication for path chase or not (PA[407], PA[410]) ' , is set at ' 1 ' , the tool path appears on the screen and the current location will be shown together. It is possible to adjust the size/color of a triangle for indicating path chase with the parameter (PA[411] ? PA[413]).
- ③ If the item, ' Whether there is an indication for path trace or not ' (PA[408], available only in View Tool Path) is set at ' 1 ' , the screen of View Tool Path will indicate the old tool paths by dot. The color and size of the trace can be adjusted at the parameter too (PA[414], PA[415]).

3. The parameter related with Preview is available under the name of ' RealTPG Screen Mode (PA[409]) ' in the item of User Parameter. If this item is set at 1, the function of preview is performed. In case of previewing too large a file created at CAM, it takes quite long time so you need to set with the parameter that the function of Preview is not performed in this case.

4. There are the function of Real TPG ' s path chase (graphic

8. Graphic Function

cursor) (PA[407]) and the parameter of path chase indication (PA[408]) ON/OFF. In addition, there is a parameter to set the size of indication (PA[411] ~ PA[414]).

5. Other parameters

- ① ‘ Real TPG Preview Stop M Code (PA[416]) ’ : The NC program, which works in loop, works in loop in Preview too so preview of the same action is continuously repeated. Therefore, you need to make out M code on the program that finishes the loop action and then, set the M code with this parameter.
- ② ‘ Selection of XC plane on TPG (PA[417]) ’ : This parameter is used when you want to show the screen on XC plane.
- ③ ‘ TPG BMP Creation Type (PA[418]) ’ : This parameter decides whether it is black/white mode or color mode when saving the path on the current graphic window as Bitmap file.(0 for black/white and 1 for color)

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PA 0363	0		Initial View Direction Setting (0-9)						
PA 0407	1		Real TPG Tracing Display Status (0:No, 1:Yes)						
PA 0408	1		Real TPG Tracing Evidence Display Status (0:No, 1:Yes)						
PA 0409	1		Real TPG Screen Mode (0:Path Display 1:Display after Preview)						
PA 0410	0		Test TPG Tracing Display Status (0:No, 1:Yes)						
PA 0411	0	Pixel	Tracer Display Width (0-20) default=10						
PA 0412	0	Pixel	Tracer Display Height (0-50) default=20						
PA 0413	0		Tracer Display Color (0:Green 1:Blue 2:Red 3:Purple 4:Cyan)						
PA 0414	1	Pixel	Tracing Evidence Display Radius Pixel (default = 2)						
PA 0415	4		Tracing Evidence Color (0:Red 1:Blue 2:Green 3:Purple 4:Cyan)						
PA 0416	0		Real TPG Preview Halt M Code (0:Not Applicable)						
PA 0417	0		XC Plane Selection on TPG (0:XYPlane, -1:-XCPlane, 1:+XCPlane)						
PA 0440	0		TPG BMP Creation Type (0:BW 1:256Color)						
PA 0441	0.0	mm	Radius of Punch Hole in NCT TPG (Default=10)						
PA 0442	0		Usage of G00 Line Display at TPG (0:Use, 1:No Use)						
PA 0443	0		Real TPG Type (0:Basic, 1:crosswise)						
PA 0444	0		Real TPG Basic Shape File Number(0:No Use)						
			DNC						

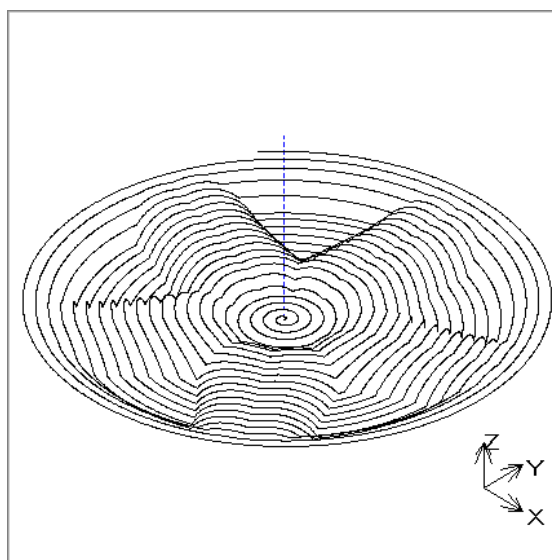
8.2 Test Tool Path(Test TPG)

This function is to check trial machining fast with graphics before actual machining. It is not the main program, but the program selected by the editor that tests. This function can perform verification faster than the test operation using MLK (Machine Lock) switch. When this function is performed, M(except for M02, M03), S, and T command are ignored.

If you test path by use of this function, you can know about the expected time for machining.

How to operate

2. Press the menu of **View Tool Path**, **F1** on the Initial Screen.
3. If it is under auto operation, the graphic window as shown on the below will appear at the left of the screen.



4. The followings are functions available at Test TPG:
 - ① Increase: Press **PgUp**. (Press the left button of mouse and drag in)
 - ② Decrease: Press **PgDn**. (Press the left button of mouse and drag

8. Graphic Function

out)

- ③ Zoom All : Press the function key of **Zoom All**, **F5**.
- ⑤ Set View Direction: Press the menu of **Set View Direction**, **F3**.
The planes set with the parameter will be shown in order. For example, if the planes are set in the order of XYZ, XY, YZ, ZX, these four views will appear in turn.
- ⑥ Move Screen (up, down, left and right): Refer to '8.1 View Tool Path'.
- ⑦ Rotation: Rotation is possible in two directions – up and down, and left and right. You can make it by use of **?**, **?**, **?**, **?**.
- ⑧ Delete Screen: Press the menu of **Delete Screen**, **F7**, and the paths that have been indicated on the screen so far will be cleared.

5. How to test TPG

- ① Press the menu of **Start Test**, **F1**, to view the whole tool paths
- ② Press the menu of **Single Block**, **F2** to view tool path for each block. The single block in the TPG test is differently carried out from that for test operation in terms of cycle code. The whole of the cycle code is performed as the concept of one block.
- ③ Press the menu of **Cancel Test**, **F6** to cancel the test in the middle of test.

Reference

For the relevant parameters, refer to '8.1 View Tool Path, 'Edit Relevant Parameters'.



Memo

9. System Management

This function is to check the system state.

9.1 Alarm List(F1)

9.2 Password(F8)

9.3 Screen of Diagnosis(F6)

9.4 Servo Graph(F5)

8. Graphic Function

9.1 Alarm List (F1)

This function is to check the history of alarms that have happened during using the system. It shows total 130 alarms that have recently happened. In case over 130 alarms have happened, the oldest alarm will disappear and the new alarm replaces it.

How to operate

1. Press the menu of **System Management**, **F6** on the Initial Screen.
2. The list of alarms that have happened so far will appear as shown on the below.

NO.	CODE	MESSAGE	DATE	TIME
20	F_80007	Cannot run the control executor.	04/10/28	00:47:26
21	F_80005	Cannot run the path generator.	04/10/28	00:51:22
22	F_80006	PLC is inoperable.	04/10/28	00:51:22
23	F_80007	Cannot run the control executor.	04/10/28	00:51:22
24	F_80005	Cannot run the path generator.	04/10/28	00:54:14
25	F_80006	PLC is inoperable.	04/10/28	00:54:14
26	F_80007	Cannot run the control executor.	04/10/28	00:54:14
27	F_80005	Cannot run the path generator.	04/10/28	09:03:19
28	F_80006	PLC is inoperable.	04/10/28	09:03:19
29	F_80007	Cannot run the control executor.	04/10/28	09:03:19
30	F_84114	SERCOS Initialization Alarm	04/10/28	09:05:01
31	F_80005	Cannot run the path generator.	04/10/28	09:05:04
32	F_80006	PLC is inoperable.	04/10/28	09:05:04
33	F_80007	Cannot run the control executor.	04/10/28	09:05:04
34	F_84011	Machine lock cannot be cancelled during automatic operation.	04/10/28	10:38:59

3. Details of each item are as follows:
 - ① Number: Showing the order of happening
 - ② Code: G_ : Alarms that have happened in PLC, F_ : Alarms that have happened in the system. For the number and details which can identify the alarms, refer to the appendix of the maintenance/repair manual.
 - ③ Contents: Messages of the alarms
 - ④ Date of happening: Date when the alarm happened
 - ⑤ Time of happening: Time when the alarm happened
4. Details of 130 alarms are displayed on the screen by 13 alarms, which are made in total 10 pages. Page change can be done by **PgUp** **PgDn**.



5. Delete Alarms: This is to remove all the alarms that have happened so far and to make it into the initial situation. Press the menu of Delete Alarm, **F2** . The function of Delete Alarm will work after the password is input.



9. System Management

9.2 Password (F8)

In order to input the parameters related with the system or to use the screen of “Diagnose” in the middle of installation/test operation, you need to input the password for the system. In the system, there are total 5 stages of password input and the contents become different, depending on the stage.

How to operate

1. Press the menu of **System Management** , **F6** on the Initial Screen, and the menu of **Password** , **F8** .
2. As shown on the below, the message of “Possible to use” or “Impossible to use” is indicated at the bottom, which depends on input of the password.

LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4
Parameter-Print Parameter-Macro State Information Servo Information Diagnosis Sequence View	Servo Drive Parameter-Axis Alarm-Delete Alarm	Recover SRAM Parameter-System Parameter-I/O Setting Parameter-HMI	
Prohibition	Prohibition	Prohibition	Prohibition

3. The password is provided in three stages and the applicable items for each stage are as shown on the above figure. The password for each stage is as follows;
 - ① 1st stage: **1111**
Possible to use the functions of parameter print, servo wave, and diagnosis.
 - ② 2nd stage: **8989**
Possible to correct axis-related parameters
 - ③ 3rd stage: **407**



Possible to correct system parameter



9. System Management

9.3 Screen of Diagnosis (F6)

It is necessary to input the password in order to get into the screen of diagnosis.

On the screen of diagnosis, various nodes in various systems (X, Y, G, F, R, T, C, D) are indicated on a real-time basis so they are used for diagnosing the system.

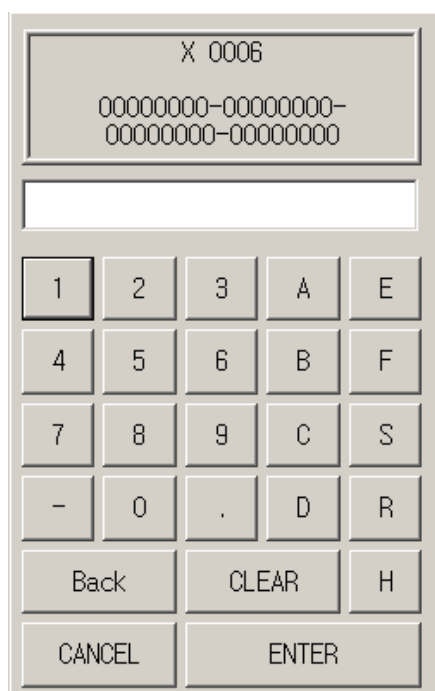
Basic structure of the screen of diagnosis

X-Y	G-F	R	T	C	D	USER			
NO.	0x18-	0x10-	0x08-	0x00	NO.	0x18-	0x10-	0x08-	0x00
X 0000	00000000-00000000-00000000-00000000	Y 0000	00000000-00000000-00000000-00000000						
X 0001	00000000-00000000-00000000-00000000	Y 0001	00000000-00000000-00000000-00000000						
X 0002	00000000-00000000-00000000-00000000	Y 0002	00000000-00000000-00000000-00000000						
X 0003	00000000-00000000-00000000-00000000	Y 0003	00000000-00000000-00000000-00000000						
X 0004	00000000-00000000-00000000-00000000	Y 0004	00000000-00000000-00000000-00000000						
X 0005	00000000-00000000-00000000-00000000	Y 0005	00000000-00000000-00000000-00000000						
X 0006	00000000-00000000-00000000-00000000	Y 0006	00000000-00000000-00000000-00000000						
X 0007	00000000-00000000-00000000-00000000	Y 0007	00000000-00000000-00000000-00000000						
X 0008	00000000-00000000-00000000-00000000	Y 0008	00000000-00000000-00000000-00000000						
X 0009	00000000-00000000-00000000-00000000	Y 0009	00000000-00000000-00000000-00000000						
X 0010	00000000-00000000-00000000-00000000	Y 0010	00000000-00000000-00000000-00000000						
X 0011	00000000-00000000-00000000-00000000	Y 0011	00000000-00000000-00000000-00000000						
X 0012	00000000-00000000-00000000-00000000	Y 0012	00000000-00000000-00000000-00000000						
X 0013	00000000-00000000-00000000-00000000	Y 0013	00000000-00000000-00000000-00000000						
X 0014	00000000-00000000-00000000-00000000	Y 0014	00000000-00000000-00000000-00000000						
X 0015	00000000-00000000-00000000-00000000	Y 0015	00000000-00000000-00000000-00000000						
X 0016	00000000-00000000-00000000-00000000	Y 0016	00000000-00000000-00000000-00000000						
X 0017	00000000-00000000-00000000-00000000	Y 0017	00000000-00000000-00000000-00000000						

For movement between items, the following functions are used same as the screen of parameter. Use F1/F2, or press the relevant TAB with mouse.

◀	Move Tab Left	Move Tab Right	Add/Delete User Item			Sequence View	System Manager	▶
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How to input data on the screen of diagnosis



The method of edit in diagnosis has been newly corrected. Go to the item to edit and press Enter (Input) key or double-click the mouse, and the new input window will appear. Here you should input values.

The method of putting data into the relevant address is as follows:

S__ : Set the bit value of 0~31 at 1. Bit should be made in hexadecimal.
 R__ : Clear the bit value of 0~31 into 0. Bit should be made in hexadecimal.
 D__ : Input the value for the relevant address in decimal.
 H__ : Input the value for the relevant address in hexadecimal.

After input is completed, press Enter (Input) key or press **ENTER** on the edit window with mouse.

9. System Management

User item on the screen of diagnosis

X-Y	G-F	R	T	C	D	USER			
NO.	0x18-	0x10-	0x08-	0x00	NO.	0x18-	0x10-	0x08-	0x00
X 0007	00000000-00000000-00000000-00000000	Y 0007	00000000-00000000-00000000-00000000						
X 0009	00000000-00000000-00000000-00000000	Y 0009	00000000-00000000-00000000-00000000						
X 0012	00000000-00000000-00000000-00000000	Y 0012	00000000-00000000-00000000-00000000						
X 0015	00000000-00000000-00000000-00000000	Y 0015	00000000-00000000-00000000-00000000						

User Item on the screen of diagnosis is used for monitoring only the necessary address items on one screen.

Locate the cursor on the address you want and press F4, and it will be automatically added to User Item. For example, if you want to monitor X5, G5 and G8, select X-T tab and press F4 on “X 0005”. Then, it will be added into User item. For G5 and G8, repeat the above.

◀	Move Tab Left	Move Tab Right	Add/Delete User Item			Sequence View	System Manager	▶
---	------------------	-------------------	-------------------------	--	--	------------------	-------------------	---

Once addition is completed, move to the right menu and User item. You will be able to confirm that the selected item has been added. In order to delete the added one, press F4 on the item in User Item that you want to delete.

The address of this User Item is cleared after HX is closed.

9.4 Servo Graph (F5)

In order to get into the screen of Servo Wave, you need to input the password of more than one stage on the screen of password.

Using the screen of Servo Wave, you can show various node status in the system through Timing Chart mode and the machine-related information through Servo Graph mode in maximum 8 graphs on a real time basis, and you will be able to use them for diagnosing the system.

1. Screen of Servo Wave

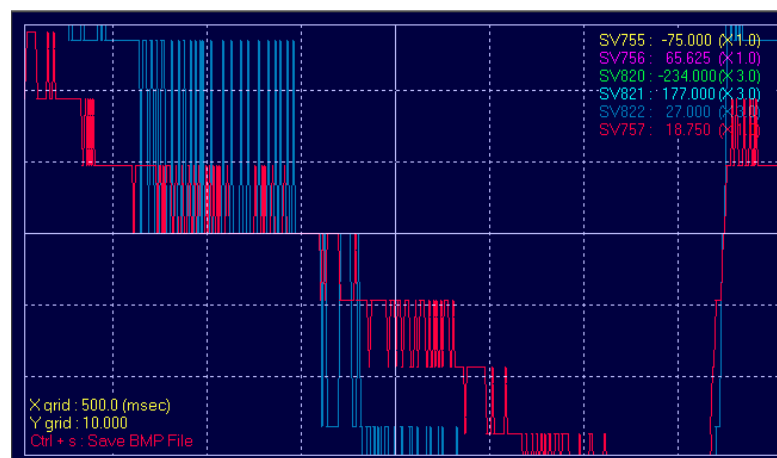
Structure of the screen

State window



This window is to deliver the functions and state appearing on the current Plot screen to users. The left-most window shows the mode of the Plot screen. The middle window shows whether servo wave starts/stops or whether file saving starts/stops. And the right-most window shows whether the functions of Trigger and Rising Time are used or not.

Plot screen



9. System Management

Plot screen shows graphs and their information to a user. Depending on the user's needs and the type of the graph, you can select either Servo Graph Mode or PLC Timing Chart Mode.

Through the Plot screen, you can know about the name and current value of the graph, grid value of X and Y, and scale.

With CUR/SEL input, the time on the cursor and the graph value relevant to the time are output on the screen.

Information related with each graph is identified by the colors shown on the Plot screen.

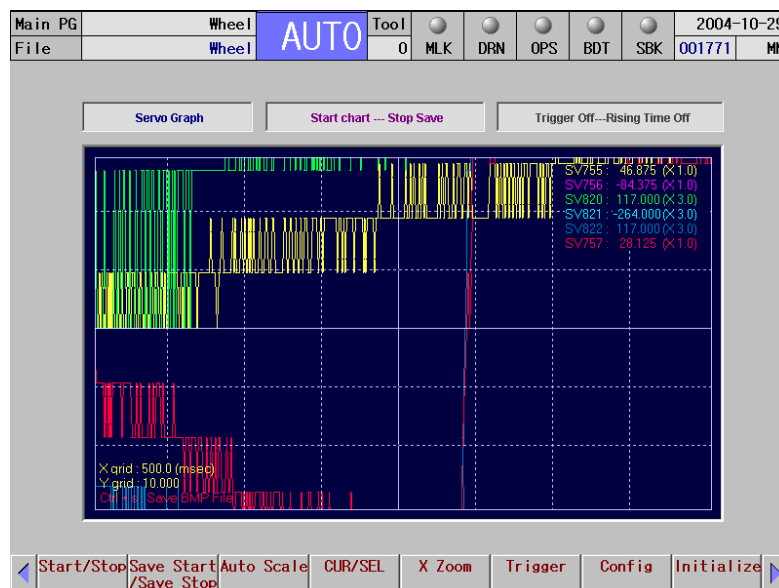
Function key

Start/Stop	Save Start/Save Stop	Auto Scale	CUR/SEL	X Zoom	Trigger	Config	Initialize
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It is to use each function of Servo Wave. Use F1/F2 or press the relevant menu with the mouse.

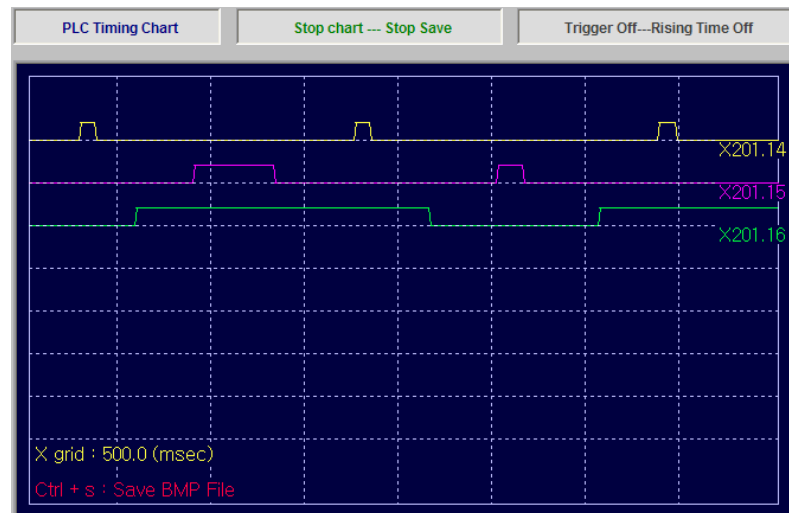
2. Description of Screen Mode

(1) Servo Graph Mode



This is used when you want to see in graph the machine-related information like axis feed, machine location of each axis, etc. and the node status in the system. It shows the input information in the similar type with oscilloscope output.

(2) PLC Timing Chart Mode



This mode is used for ON/OFF of PLC sequence or for seeing timing of the change in the nodes in the system. The graph divides bit and constant data so it is displayed by ON/OFF signal or data signal as shown on the figure. In case of data signal, the value is output on the graph. At the right bottom of the graph, the name of the graph appears.

The functions of Auto Scale and of adjusting scale of Y are not used.

3. Description of functions

Start/Close (F1)

This is used to start or close output of graphs. It can be checked through the state window. When starting, all the settings become initialized with the contents in the Config dialog box.

9. System Management

Start Save/Close Save (F2)

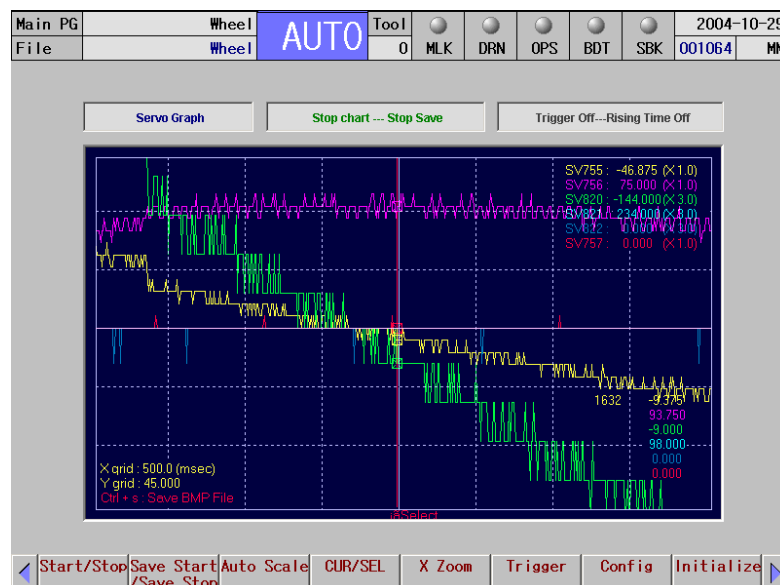
This is to save the graph data in a file or to stop to save. The state window tells whether the file is saved or the saving is stopped. The file is saved under HX folder.

Auto Scale (F3)

It automatically adjusts the scale of the graph appearing on the screen suitably for the size of the screen. If you input the menu of Auto Scale, the contents of Y Zoom Step in Config dialog box is changed into 1/5.

It is not used during PLC Timing Chart Mode

CUR/SEL (F4)



This is used to check the value of a particular part on the screen. If you select CUR/SEL by use of a function key, red line and box will appear on the Plot screen. Move the red line, using $\leftarrow \rightarrow$, and the box will move, flowing the graph. At this time, the graph value in the part where the box is located is shown at the right bottom of Plot screen.

X Zoom (F5)

It is used to expand and see a particular part on the Plot screen.

Using the function key, select CRU/SEL, and using $\longleftrightarrow$, go to the part that you want. This is the function of CUR/SEL. Here, select CUR/SEL again, and “Select” will appear below the current location of the cursor. And then, move the cursor as much as needed and press X-Zoom button. Then, the graph between where you selected first and the current location of the cursor is expanded to appear on the Plot screen. Press CUR/SEL again, and the expansion of the part is cancelled.

Config (F6)

ch	Map Name	Number	Start Bit	End Bit	Scale	Use
1	SV	755	0	0	0	☒
2	SV	756	0	0	0	☒
3	SV	820	0	11	11	☒
4	SV	821	0	11	11	☒
5	SV	822	0	11	11	☒
6	SV	757	0	0	0	☒
7	None	0.0	0	0	0	☐
8	None	0.0	0	0	0	☐

If Config key is input, the dialog box like the figure will appear. Through the dialog box, you can input/adjust the information on the graph and the type of the map that you want to show on the Plot screen. The contents put into the Config dialog box are saved under the name of Config.txt in the folder where HX is installed.

Mode Select

9. System Management

You decide the mode to output the graph, either Server Graph Mode or PLC Timing Char Mode.

X Grid

You decide the X grid value of the graph which will be output on the Plot screen.

Y Grid

You decide the Y grid value of the graph which will be output on the Plot screen.

Y Zoom Step

According to input of  , change in the scale adjustment is decided. If you press  , the scale is adjusted as much as the input value.

Map Name

Input the name of the map to output

Number

Input the number of the map to output

Start Bit

Input this when you want to print a particular part of the map. Input the start bit address of the part in hexadecimal you want.

End Bit

You input this when you want to print a particular part of the map. Input the end bit address of the part in hexadecimal you want.

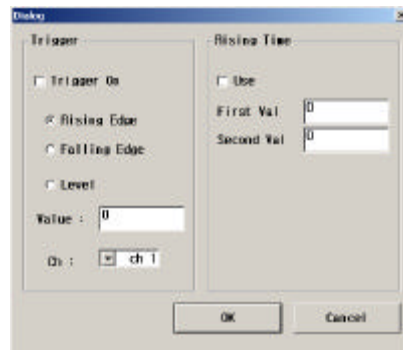
Scale

Input scale of each channel. The value increased as much as input value is shown on the Plot screen.

Use

Decide whether to output graph or not.

Trigger (F7)



If you input the button of Trigger, the dialog box as shown on the below figure will appear. Input whether to use Trigger and Rising Time functions, together with necessary values.

Whether to use Tirgger

Decide whether to use Trigger function or not. If Trigger is selected, when the graph passes through the input particular value, the graph temporarily stops under each condition.

Rising Edge, Falling Edge, Level

You can select one of these.

Value

Input the value when using the function of Trigger.

Ch

Decide the channel to which Trigger is applied.

Whether to use Rising Time

Decide whether to use the function of Rising Time. If Rising Time is selected, when the graph passes through the middle of two values whose order is determined, the time difference is shown on the screen.

First Val

Input the first value when using the function of Rising Time.

Second Val

Input the second value when using the function of Rising Time.

Initial Status (F8)

9. System Management

It initializes the setting for the Plot screen. If Auto Scale key is input, the scale of Y doesn't change

Scale Adjust

You can adjust the scale of the graph by use of . The current scale of the graph appears at the left bottom of the Plot screen. It is not used under PLC Timing Chart Mode.

Move

You will be able to move the graph left and right by use of ? , ? . At this time, the graph should be stopped temporarily.



Memo

10. Edit Parameter

In this chapter, setting and operation of the parameters for HX system are described.

10.1 Structure of Parameter

10.2 Class of Parameter

10.3 Find Parameter

10.1 Structure of Parameter

The screen of Edit Parameter is divided into user-related parameter and system-related parameter. The system-related parameter is protected by password in order to prevent wrong change by user's mistake.

1) The user-related parameter consists of:

Program, User, Machining 1 and Machining 2

2) The system –related parameter consists of:

System, Macro, Axis, I/O Setting, Special Functions, and HMI Setting

Basic structure of Parameter screen

10. Edit Parameter

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
*PP 0020	0		TPG Applicable Equipment Type (0:Milling 1:Lathe 2:NCT 3:Point)						
			Axis Display Setting						
*PA 0406	0		Automatic Axis Display Usage Status (0:Use, 1:No Use)						
PA 0330	0.0		Format (Total Number of Digits/Floating Digits) Default: 10.3						
PA 0329	0		Radius/Diameter Display Setting (0:Radius, 1:Diameter)						
PA 0331	0		Radius/Diameter Display Function Usage Status per Axis (0:No, 1:Apply) (1						
PA 0332	0		Radius/Diameter Display Function Usage Status per Axis (0:No, 1:Apply) (2						
PA 0333	0		Radius/Diameter Display Function Usage Status per Axis (0:No, 1:Apply) (3						
PA 0334	0		Radius/Diameter Display Function Usage Status per Axis (0:No, 1:Apply) (4						
PA 0000	0		Display Unit (0:Metric, 1:Inch)						
PA 0374	0		Inch/Metric Usage Status per Axis (0:No Application 1:Apply) (1 Axis)						
PA 0375	0		Inch/Metric Usage Status per Axis (0:No Application 1:Apply) (2 Axis)						
PA 0376	0		Inch/Metric Usage Status per Axis (0:No Application 1:Apply) (3 Axis)						
PA 0377	0		Inch/Metric Usage Status per Axis (0:No Application 1:Apply) (4 Axis)						
*Ax 0000			Axis Name (Max. 2) (1 Axis)						
*Ax 0001			Axis Name (Max. 2) (2 Axis)						
*Ax 0002			Axis Name (Max. 2) (3 Axis)						
*Ax 0003			Axis Name (Max. 2) (4 Axis)						

For movement between items (TAB), use the mouse or F1 or F2. In addition, you can check the parameter at the bottom of the screen by use of PageUp/pageDown.

Details of all the parameters such as number, value, unit and description are indicated in a line so a user can understand easily.

Put the cursor on the parameter to correct and input a new parameter value. The gray part means that the password is not input and that they are impossible to input.

In order to move the items of the parameter, use F1 (left) or F2 (right) as shown on the below figure. v

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
			General Setting						
PA 1431	0	Count	Target Machining Count						
SN 0101	0	Count	Machining Count						
PI 0072	0		Tool Diameter Compensation Method (0:Diameter 1:Radius)						
PI 0073	1		X Axis Command Method (0:Diameter 1:Radius)						
PI 0076	1		Decimal Point Inspection (0:Yes 1:No)						
PI 0170	0.000	mm	Min. Input Increment(Default:0.001),Used for Decimal Point Inspect						
PI 0082	0		Perpendicular Chamfering Method (0:UK 1:C Instruction)						
PI 0120	0		Dwell Command Method (0:Time 1:Revolutions)						
PI 0128	0		Tool Diameter Compensation Type (0:Detour 1:Direct)						
PI 0172	0		Checking the Tool interference at the Tool compensation(0:No 1:Yes)						
PI 0133	1		Block Stop at reset (0:Maintain, 1:Initial Block, 2:Called Block)						
PI 0134	0		Sequence number search in cycle(0:Yes 1:No)						
PI 0151	0.000	mm	Radius Tolerance Deviation of circular block						
PI 0171	0		Regenerating a center of arc when a arc alarm occurred(0:No 1:Yes)						
PI 0155	0		Rotating Axis of cylinder interpolation(0:X 1:Y 2:Z)						
PI 0177	0.000	mm	Allowable NURBS Interpolation Error						
PI 0178	0.000	mm	NURBS Interpolation Tool Radius						



PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PA 0363	0		Initial View Direction Setting (0-9)						
PA 0407	1		Real TPG Tracing Display Status (0:No, 1:Yes)						
PA 0408	1		Real TPG Tracing Evidence Display Status (0:No, 1:Yes)						
PA 0409	1		Real TPG Screen Mode (0:Path Display 1:Display after Preview)						
PA 0410	0		Test TPG Tracing Display Status (0:No, 1:Yes)						
PA 0411	0	Pixel	Tracer Display Width (0-20) default=10						
PA 0412	0	Pixel	Tracer Display Height (0-50) default=20						
PA 0413	0		Tracer Display Color (0:Green 1:Blue 2:Red 3:Purple 4:Cyan)						
PA 0414	1	Pixel	Tracing Evidence Display Radius Pixel (default = 2)						
PA 0415	4		Tracing Evidence Color (0:Red 1:Blue 2:Green 3:Purple 4:Cyan)						
PA 0416	0		Real TPG Preview Halt M Code (0:Not Applicable)						
PA 0417	0		XC Plane Selection on TPG (0:XYPlane, 1:XCPlane, 1:+XCPlane)						
PA 0440	0		TPG BMP Creation Type (0:BW 1:256Color)						
PA 0441	0.0	mm	Radius of Punch Hole in NCT TPG (Default=10)						
PA 0442	0		Usage of G00 Line Display at TPG (0:Use, 1:No Use)						
PA 0443	0		Real TPG Type (0:Basic, 1:crosswise)						
PA 0444	0		Real TPG Basic Shape File Number(0:No Use)						
			DNC						

◀	Move Tab Left	Move Tab Right	Move Axis Down	Move Axis Up	Find Parameter	Print	▶
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10.2 Class of Parameter

Program-related parameter

Parameters for making and preparing NC program required for auto operation are available here.

User parameter

There are TPG-related parameter for which a user can change the setting if necessary, and DNC communication parameter.

Machining 1 and Machining 2 parameter

There are parameters that are used for manual operation and for auto operation related with machining.

System parameter

This is related with system setting. It is for setting control axis and H/W structure environment.

Macro parameter

This is related with system macro. It sets whether to edit macro program or not and also sets the number of paging G code/M code macro programs.

Axis parameter

This is for setting the parameter related with servo and spindle that are used in the system. It supports maximum 32 axes and for each axis, various parameters can be set.

I/O setting parameter

This is for setting the parameters related with In/Out module. It sets ID of I/O module and the starting address of I/O that are used in the system.

Special function parameter

For the machines to which special functions are added, you can set the parameters here. For example, in case of laser cutter, Z gap trace function is used and you can set various parameters in relation to the characteristics and signal handling of the sensor.

In case of quilting machine, you are able to set the parameters related with stitch function and the parameters related with the screen of pattern select.

HMI parameter

This is for setting the parameters related with HMI (Human Machine Interface) of the system. You can set the descriptive name of the indicative unit system and axis, yes/no of axis indication, etc.

Ex) Basic shape of the screen of "Axis" Parameter

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	1 AXIS
NO.	Value	Unit	Comment						
			Servo Spindle Common Parameter						
*PS 0001	1		Axis Type Setting (0:No Use, 1:Servo, 2:Spindle)						
*PS 0002	1		Axis Port No.						
*PS 0003	1		CNC_Drive Interface Type (0:SERCOS, 1:Analog)						
			Servo Axis Setting						
*PS 0021	1		Axis Motion Type (1:Linear(V),2:Rot (V),3:Linear(P),4:Rot. (P))						
*PS 0025	0		Drive Origin Return Usage (0:No Use, 1:Use)						
*PS 0032	0		Master Axis Number, if slave axis in synch. control						
*PS 0044	0.0000	mm,deg	Backlash Amount						
PS 0047	0.000	Voltage	Friction Compensation Value						
*PS 0053	1		Pitch Error Compensation Usage(0:Compensate,1:No Compensation)						
*PS 0057	0.0000	mm,deg	Pitch Error Input Interval						
*PS 0058	0		Pitch Error Table Index corresponding to Origin						
*PS 0259	0	msec	Position Type Servo Miss Step Compensation Value						
*PS 0260	0	pps	Position Type Servo Remaining Pulse Compensation Value						
*PS 0269	1		Number of Machine-Side Gear Teeth						
*PS 0270	1		Number of Motor-Side Gear Teeth						
*PS 0271	5.0000	mm	Ball Screw Pitch						

It is possible to set maximum 32 axes. In order to find an axis in order, use F3/F4. The axis number of the parameter that is selected is shown on the right top of the screen of Edit Parameter.

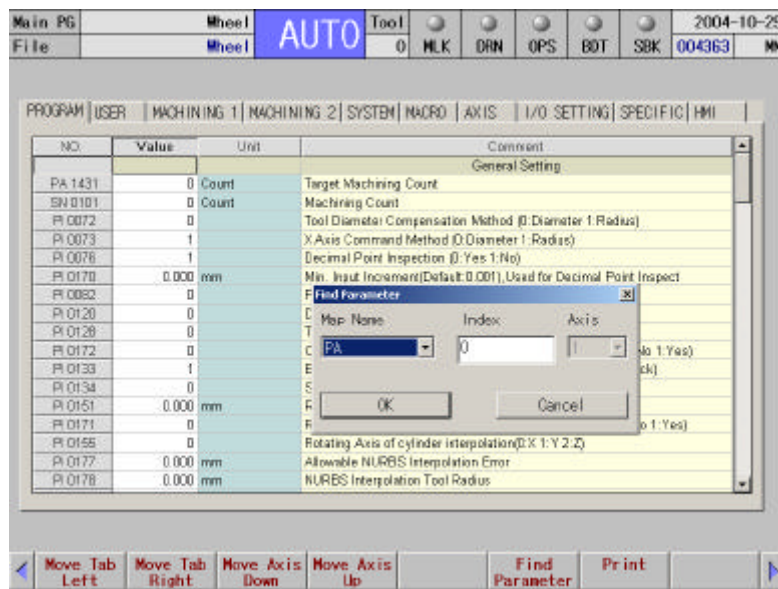
10. Edit Parameter

PROGRAM		USER		MACHINING 1		MACHINING 2		SYSTEM		MACRO		AXIS		I/O SETTING		SPECIFIC		1 AXIS	
NO.	Value	Unit	Comment																
Servo/Spindle Common Parameter																			
*PS 0001	1		Axis Type Setting (0:No Use, 1:Servo, 2:Spindle)																
*PS 0002	1		Axis Port No.																
*PS 0003	1		CNC_Drive Interface Type (0:SERCOS, 1:Analog)																
Servo Axis Setting																			
*PS 0021	1		Axis Motion Type (1:Linear(V),2:Rot.(V),3:Linear(P),4:Rot.(P))																
*PS 0025	0		Drive Origin Return Usage (0:No Use, 1:Use)																
*PS 0032	0		Master Axis Number, if slave axis in synch. control																
*PS 0044	0.0000	mm,deg	Backlash Amount																
PS 0047	0.000	Voltage	Friction Compensation Value																
*PS 0053	1		Pitch Error Compensation Usage(0:Compensate,1:No Compensation)																
*PS 0057	0.0000	mm,deg	Pitch Error Input Interval																
*PS 0058	0		Pitch Error Table Index corresponding to Origin																
*PS 0259	0	msec	Position Type Servo Miss Step Compensation Value																
*PS 0260	0	pps	Position Type Servo Remaining Pulse Compensation Value																
*PS 0269	1		Number of Machine-Side Gear Teeth																
*PS 0270	1		Number of Motor-Side Gear Teeth																
*PS 0271	5.0000	mm	Ball Screw Pitch																

Move Tab Left	Move Tab Right	Move Axis Down	Move Axis Up	Find Parameter	Print
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10.3 Find Parameter

You can go to the parameter that you want on the screen of Edit Parameter.
Press 'F6 Find Parameter', and the dialog window as shown on the below will appear.



1. Use $\uparrow \downarrow$ to decide the kind of parameter on the Map Name window that you want.
2. Press Tab to move to Index window. You should input the parameter number here, and then, input ENTER..
3. If it is axis-related parameter PS map, you should select the axis number (1 ~ 32) together.

Special notes

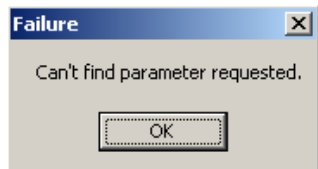
1. This function works only for the parameters registered with the Parameter screen. For example, for the offset value of the work-piece coordinate system available on the screen of Coordinate System Setting, this function doesn't

10. Edit Parameter

work.

2. In case it is impossible to find the parameter on the Parameter screen, the following message window will appear.

3. The parameter that is not available on the screen of Set Parameter can't be found by this function (ex: offset value of the work-piece coordinate system on the Setting screen).







Memo



11. Utility

This is explanation of the convenient utility that is used in HX.

11.1 DNC

11.2 Scheduling



11.1 DNC

With this function, you can give and receive a program with other PC, by use of RS232C serial port, COM1.

1. Set Communication Parameter

Communication parameter can be set in the User Item of parameter editor (F4 in the Initial Screen).

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
NO.	Value	Unit	Comment						
PA 0417	0		XC Plane Selection on TPG (0:XYPlane, -1:-XCPlane, 1:+XCPlane)						
PA 0440	0		TPG BMP Creation Type (0:BW 1:256Color)						
PA 0441	0.0	mm	Radius of Punch Hole in NCT TPG (Default=10)						
PA 0442	0		Usage of G00 Line Display at TPG (0:Use, 1:No Use)						
PA 0443	0		Real TPG Type (0:Basic, 1:crosswise)						
PA 0444	0		Real TPG Basic Shape File Number(0:No Use)						
			DNC						
*PP 0002	1		Communication Port (0:COM1, 1:COM2, 2:TCP/IP)						
PI 0001	0	bps	Baudrate (1200,2400,4800,9600,19200) (#1)						
PI 0002	0	bps	Baudrate (1200,2400,4800,9600,19200) (#2)						
PI 0003	0		Parity Bit (0:None, 1:Odd, 2:Even) (#1)						
PI 0004	0		Parity Bit (0:None, 1:Odd, 2:Even) (#2)						
PI 0005	0		Data Bit (#1)						
PI 0006	0		Data Bit (#2)						
PI 0007	0		Stop Bit (#1)						
PI 0008	0		Stop Bit (#2)						
PI 0009	0		EOT Code (0:None, 1:0x03, 2:0x04, 3:0x14) (#1)						
PI 0010	0		EOT Code (0:None, 1:0x03, 2:0x04, 3:0x14) (#2)						

- Communication Port: This is to give and receive data. You can designate COM1 and COM2. After you change these values, you have to restart the system necessarily.
- Baudrate: This to set the communication speed, which is available from 1200 to 19200. When you connect, you have to make it same as the setting for the opposite equipment. In general, 9600 is used.
- Parity Bit: This is to set the signal that checks errors in the contents of the transmitted data. In general, Even is used for the communication using ISO code.
- Data Bit: This is to set the length of the transmitted data. In general, 7 bits is used for the communication using ISO code.
- Stop Bit: This is to set the stop signal that is added next to one transmitted data. In general, 1bit is used.

- EOT Code: Abbreviation of End Of Transmission, it is to set the control signal that informs the receiver of the closing signal of the transmitted data. When receiving it, it automatically closes upon receipt of the signal. In general, 0x14 (DC4) is used.
- The items that are basically defined, even though they are not shown in the item of parameter, are as follows:
 - ① FLOW CTRL: X-ON/X-OFF is used for flow of the data that is input/output. (At the time of data input/output, software control by DC1 and DC3 communication code makes communication)
 - ② For EOB (End of Block) that informs the end of each block for input/output data, LF is used.
 - ③ There is no waiting for transmission. When receiving, it sends X-ON code of the sender.

2. Transmission

1. The screen of DNC is as follows:

<Basic screen of DNC>

The screenshot displays a software interface for DNC (Direct Numerical Control) with two main sections. The top section, titled 'File Name', contains a text input field for the file name and a 'Size (Byte)' field with a numeric input. Below this is a 'Path' field with a text input. The bottom section, titled 'Status', contains a 'Status' field with a dropdown menu, a 'Size (Byte)' field with a numeric input, a 'Rate (%)' field with a numeric input, and a 'Block' field with a text input.

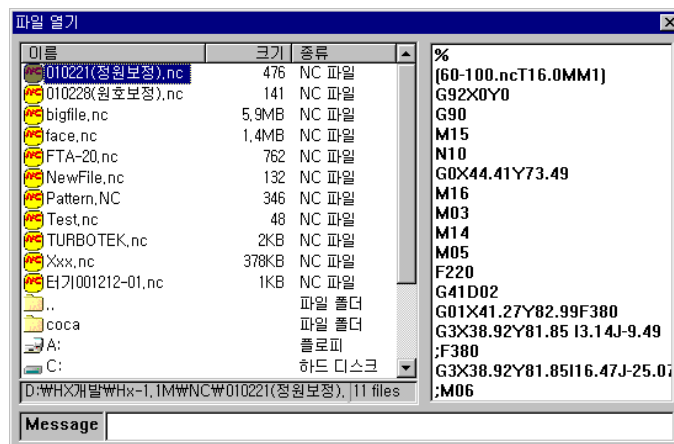
- ① The window of 'Communication State' indicates 'In transmission' when data is being transmitted.
- ② The window of 'File Path' indicates the location of the folder for the selected file.

- ③ The window of 'File Name' indicates the name of the transmitted file.
- ④ The window of 'File Size' indicates the size of the transmitted file.
- ⑤ The window of 'Transmission Rate' indicates the percentage (the amount of data that is output during transmission is divided by the file size) by second.
- ⑥ The window of 'Block' indicates the number of accumulated blocks for data that is output during transmission in the unit of 1 second.
- ⑦ The window of 'Transmission Amount' indicates the amount of accumulated data that is output during transmission in the unit of 1 second.
- ⑧ The window of 'Output Details' indicates the data under output in the unit of block. The output data can be different from the shown data because of the difference in speed.

<DNC Initial Men>



2. Press **F3**.



3. The screen of File Select as shown on the below appears. At this time, the folder (path of file) is the location set by the system, and it is 'NC' folder under the folder in which the system is installed. On the screen

of File Select, it is possible to change the location of the folder. After change, if you want to change to the default folder location immediately, you should select 'NC DIR' which is available in every folder

통신 상태	송신 중...
파일 경로	D:\HX개발\HX-1.1M\NC#
파일 이름	1222.nc
파일 크기 [Byte]	2631
전 송 륜 [%] [38]	
블 록	134
전 송 량 [Byte]	1024
G54	

4. Select the program to transmit and then, double-click the mouse or press **ENTER** .
5. The information on the selected file is shown on the below figure.
6. The menu is changed as shown on the following. For transmission, press the start button, **F5** . In order to cancel the transmission, press **F8** . (Once started, it doesn't wait for X-ON signal and outputs immediately.)

D N C	Send Program	Receive Program	Schedule Program
-------	--------------	-----------------	------------------

7. The file is output through serial port (port designated by parameter / refer to 'Set Communication Parameter'), and the progress details like the following appear. Press **F8** in order to stop during operation. If an error happens, you should check setting for communication parameter.

파일 경로	D:\HX개발\HX-1.1M\NC\
파일 이름	1222.nc
파일 크기 [Byte]	2631

8. Once all the files are output, it automatically closes and returns to the initial menu state.

3. Receiving

1. DNC screen is as follows;

<Basic screen of DNC>

- ① The window of 'Communication State' indicates 'In receiving' when data is being received.
- ② The window of 'File Path' indicates the location of the folder in which the input file is saved.
- ③ The window of 'File Name' indicates the name of the received file.
- ④ The window of 'File Size' indicates 0 because the size of the received file is unknown in advance.
- ⑤ The window of 'Transmission Rate' indicates nothing because the size of the received file is unknown in advance.
- ⑥ The window of 'Block' indicates the number of the accumulated blocks for data that is input during receiving in the unit of 1 second.
- ⑦ The window of 'Transmission Amount' indicates the amount of accumulated data that is input during receiving in the unit of 1 second.
- ⑧ The window of 'Output Details' indicates the receiving data in the unit of block. Because of the difference in speed, the received data can be different from the shown data.

2. Press **F3**.

3. The location of the cursor moves to the window of the name of the file to receive.

<Figure>

4. Input the name of the file to receive and press **ENTER**. The location of the folder (file path) becomes the folder location that is changed on the screen of Select File to Transmit
5. Press **F5** to start to receive. Press **F8** to cancel. (Once started, the signal of X_ON is output once.)

<Figure>

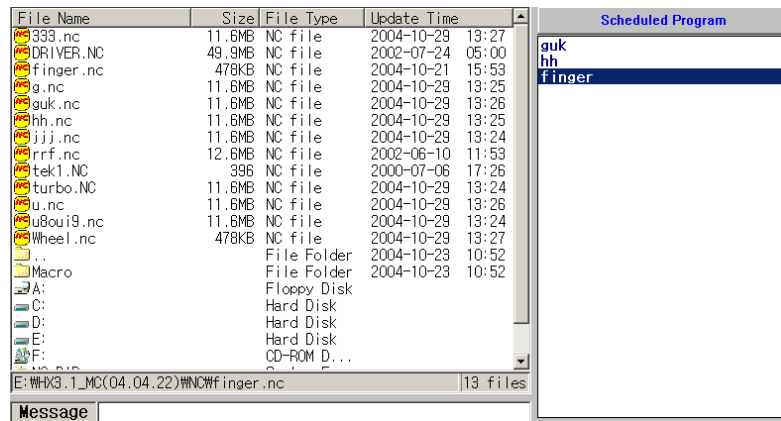
6. Once all the input data come in through serial port (port designated by the parameter / refer to 'Set Communication Parameter') and EOT code designated by parameters (EOT designated by parameter / refer to 'Set Communication Parameter') is input, it automatically closes and returns to the initial menu status. If an error happens, you should check the setting for communication parameter.

11.2 Scheduling

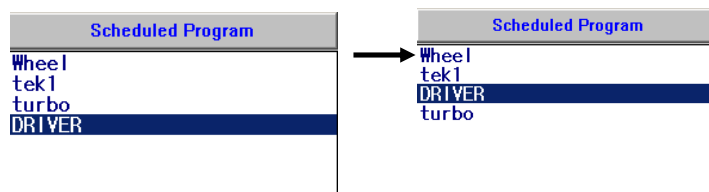
This function makes unmanned operation possible by scheduling the programs to process in advance. For more details, refer to 'Chapter 3. Auto Operation'.

How to operate

1. Basically, the following screen appears. Select the program for scheduling and then, double-click the mouse or press **ENTER** on the keyboard or MDI key, and scheduling program will be added. Once the program is selected, it is added under the scheduling program list at the right of the screen.



2. As per the above 1, continuously add the scheduling programs. If you want to change the order of scheduling the programs, use **F6** **F7** to move the cursor to the program that you want and then, use **F4** **F5** to change the order.



3. If setting for scheduling is completed, press the menu of Schedule Set/Cancel, **F8** to start the mode of scheduling process (Press **F8** to cancel scheduling too)
4. In the mode of scheduling process, the message in the below blinks at the bottom of the screen.

F_92012[1/ 1] : Scheduling production mode.

5. In the mode of scheduling process, if you do Cycle Start, scheduling process gets started.



Memo





12. Screen Composition

- 12.1 Screen Composition**
- 12.2 Common Screen**
- 12.3 Initial Screen**
- 12.4 View Toolpath (F1)**
- 12.5 Setting (F2)**
- 12.6 Edit (F3)**
- 12.7 Parameter (F4)**
- 12.8 Utility (F5)**
- 12.9 System Management (F6)**
- 12.10 Program Management (F8)**
- 12.11 Screen Help Function (SHIFT + F12)**



12. Screen Composition

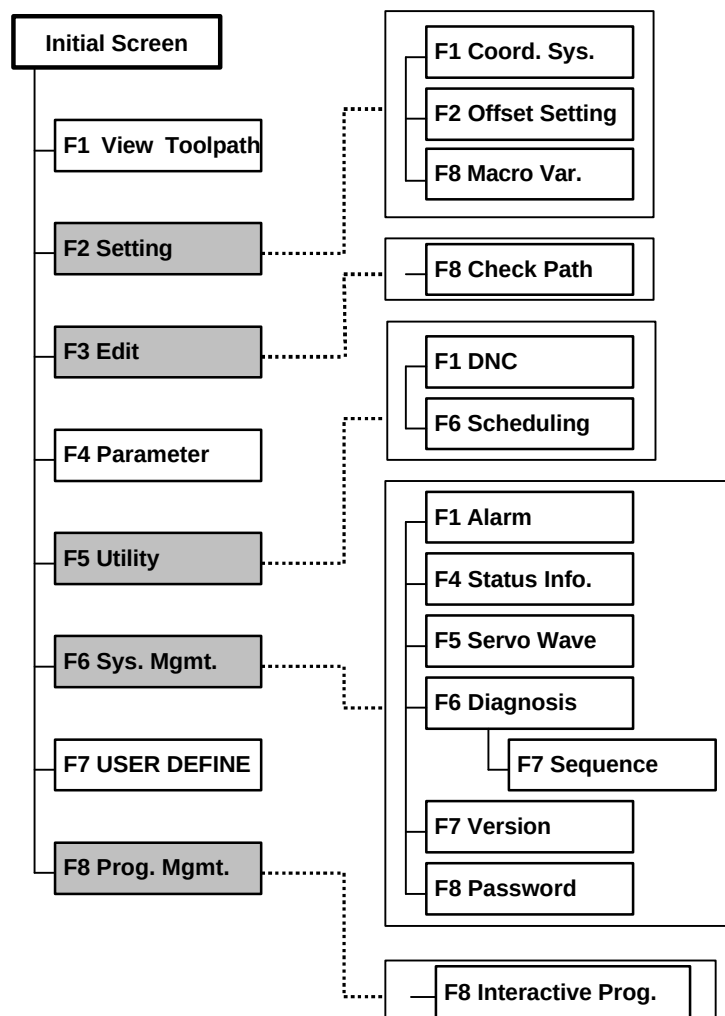
12.1 Screen Composition

Use ? (F9) key leftmost of function keys to return to the initial screen and menu and use ? (F10) key rightmost of function keys to move to the previous screen.

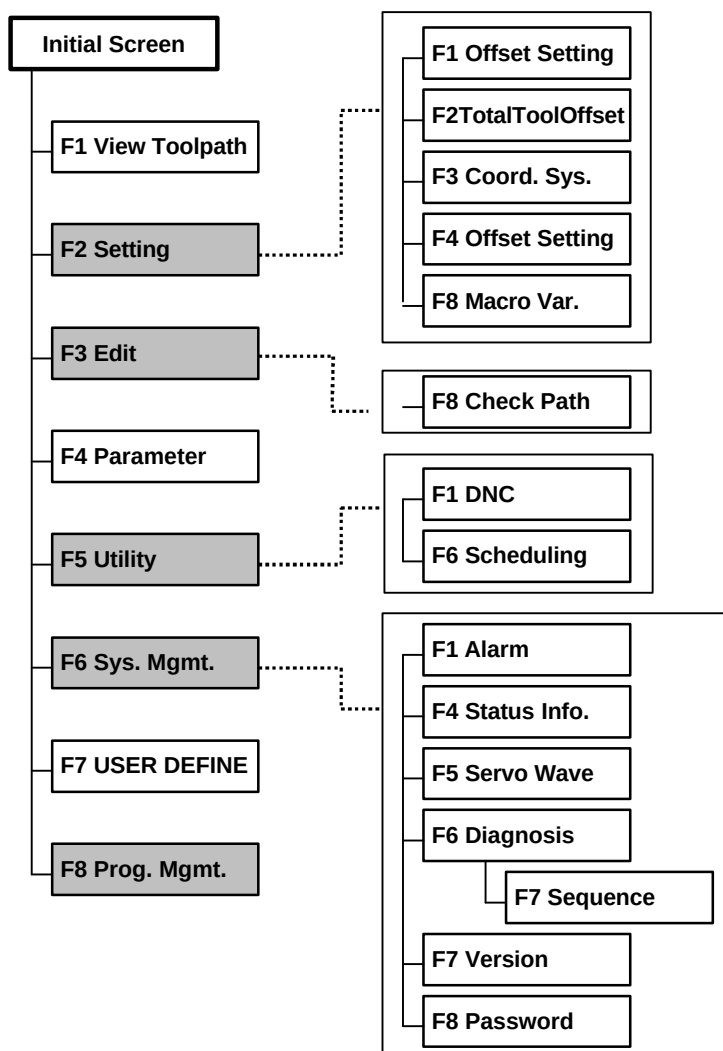
However, MDI related screen is shown as the initial screen in MDI Mode and the position display screen is shown in DNC Mode.

1. Machining Center Screen

Left side shows the main menu items and right side shows the submenu items that are available when the corresponding main menu item is selected.



2. Turning Center Screen (Lathe)



12. Screen Composition

12.2 Common Screen

1. Common screen is located at the top of HX-Series screen. It displays the basic information including the current program, tool number, operation mode, control panel setting status, date, and default coordinate system.

Main PG	Wheel	JOG	Tool							2004-10-22
File	Wheel		0	MLK	DRN	OPS	BDT	SBK	000000	MM

2. Machining Program Name

Main PG	Wheel
File	Wheel

Main PG : Name of the selected main program.

Current PG : Name of the machining program currently running. When a sub-program is running, the name of that sub-program is shown.

3. Current Mode



Displays the operation mode. There are DNC, EDIT, AUTO, MDI, JOG, MPG, STEP, ZRN operation modes available.

4. Current Tool Number and Control Panel Status Display

Tool					
0	MLK	DRN	OPS	BDT	SBK

Displays the current Tool Number and control panel status information. If the control panel switch is ON, it is shown in red.

5. Date, Machining Line Number, & Units

2001- 8- 3
000000 MM

Displays the current date, program line number and unit system (MM/Inch).

12.3 Initial Screen

The following 2 screens are available as the initial screen (? , F9). Page Down key can be used to change the screen. In this initial screen, you can check various machining condition information such as the currently running AUTO Mode program, various coordinates, operation time, machining feed, and spindle revs.

? Initial Screen #1.

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									000010 MM

1 / 2

00001		Absolute		Machine	
G17G40G80		X	0.048	X	0.048
S500M03		Y	0.005	Y	0.005
G90G54X0.000Y0.000Z50.000		Z	3.600	Z	3.600
G00 X0.048Y0.005					
Z-5.633					
G01F100				Distant To Go	
X0.094Y0.019		X	0.000	X	0.000
X0.137Y0.042		Y	0.000	Y	0.000
X0.176Y0.074		Z	0.000	Z	0.000
X0.210Y0.114					
X0.236Y0.162					
X0.256Y0.215					
X0.266Y0.274					
X0.267Y0.337					
X0.258Y0.402					
X0.238Y0.468					
X0.208Y0.534					
X0.166Y0.598					
X0.114Y0.659					
X0.051Y0.714					

FEED	0.0	OVERRIDE	50
SPINDLE	0	OVERRIDE	50
Time	15:02:13	Oper. Time	0:40:59
Cyc. Time	0:00:03	T. Cyc. Time	0:00:03
DWELL	0	Count	0

L 00010 X0.176Y0.074

Tool Path Graphic	Setting	Edit	Parameter	Utility	System Manager	Program Manager
-------------------	---------	------	-----------	---------	----------------	-----------------

- You can check the currently running machining program. At the bottom, machining block is displayed.
- Various coordinates (current position of the target object coordinate system, machine position and remaining distances)
- Various system-information can be checked, such as Machining Feed, Spindle Revs, Overrides, and various time displays.
- When you press the leftmost of soft keys, ? (F9), it is switched to the initial screen #1.

12. Screen Composition

? Initial Screen #2.

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									000010 MM

2 / 2

Absolute		Machine		Relative	
X	0.048	X	0.048	X	0.048
Y	0.005	Y	0.005	Y	0.005
Z	3.600	Z	3.600	Z	3.600

Distant To Go		Handle Interrupt	
X	0.000	X	0.000
Y	0.000	Y	0.000
Z	0.000	Z	0.000

Time	15:43:50	Cyc. Time	0:00:03
FEED	0.0	D	0
SPINDLE	0	H	0

G00 G97 G90 G-1 G94 G21 G40
 G-1 G22 G80 G50 G67 G49 G54
 G64 G17 G15 G69 G98 G113 G11.3

F_92012[1/ 1] : Scheduling production mode.

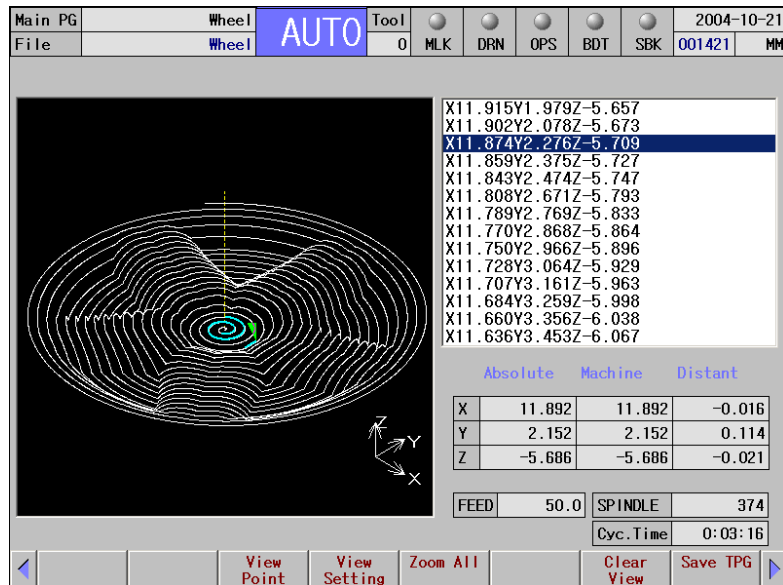
Tool Path	Setting	Edit	Parameter	Utility	System Manager	Program Manager
-----------	---------	------	-----------	---------	----------------	-----------------

- Various coordinates (current position of the target object coordinate system, machine position, remaining distances, relative position, and the amount of handle interrupt)
- Various system-information can be checked, such as Machining Feed, Spindle Revs, Overrides, and various time displays.
- Modal information display of G code group.
- This screen is selected by pressing Page Down key from the initial screen #1.
- This is also the initial screen for DNC Mode.

12.4 View Tool Path (F1)

? View Tool Path screen (F1)

This View Tool Path screen displays the AUTO Operation tool path in real time.



- You can see the tool path for actual machining. Also, you can check the current position and machine position.
- **View Direction Setting F3** key lets you change the viewing position.
- **View Center Setting F4** key allows you to redefine the screen center.
- **Zoom All F5** key zooms in/out the tool path to fit in the screen.
- **Clear F7** key erases the tool path currently displayed in the screen.
- **Save Tool Path F8** key allows the storage of the tool path shown in the screen to a separate image file.
- For more information, refer to 'Chapter 8. Graphic Function'.

12. Screen Composition

12.5 Setting (F2)

In this Setting screen, you can set various options such as coordinate system offset of numerous target objects, tool compensation, tool wear-out compensation, and global macro variable. For more information on offset setting, refer to 'Chapter 13. Setting'.

? Tool Shape Compensation Setting Screen. (Lathe)



- While checking the position information, you can set the shape compensation for each tool, coordinate system shift, and relative coordinates.
- Settings like Tool **인선** Shape and **인선** Radius corresponding to Tool Offset Number of total 64 tools can be made.

? Total Tool Offset Setting Screen. (Lathe)

메인PG	3002	JOG	공구	0	MLK	DRN	OPS	BDT	SBK	2001- 9-24
전행PG	3002									000000

인선형상번호 (0-9)

기계위치 X -148.012
Z -176.000

상대위치 X -148.012
Z -176.000

FEED	0.0	OVER.	100
SPINDLE	0	OVER.	100

작업물좌표계	취소	좌표계 이동	사용
		Z축 Shift	2.000

음셋	인선	GX	GZ	WX	WZ	RAD
01	3	-200,000	-180,000	0,000	0,000	0.80
02	3	-200,000	-180,000	0,000	0,000	0.80
03	3	-200,000	-180,000	0,000	0,000	0.40
04	0	0,000	0,000	0,000	0,000	0,00
05	0	0,000	0,000	0,000	0,000	0,00
06	0	0,000	0,000	0,000	0,000	0,00
07	0	0,000	0,000	0,000	0,000	0,00
08	0	0,000	0,000	0,000	0,000	0,00
09	0	0,000	0,000	0,000	0,000	0,00
10	0	0,000	0,000	0,000	0,000	0,00
11	0	0,000	0,000	0,000	0,000	0,00
12	0	0,000	0,000	0,000	0,000	0,00

- Using the machine position and relative position information, the shape compensation (GX, GZ) for each tool, wear-out compensation (WX, WZ), and tool 인선 shape number and 인선 radius values corresponding to each tool offset number can be set.
- Settings like Tool 인선 Shape and 인선 Radius corresponding to Tool Offset Number of total 64 tools can be made.
- Absolute offset input method and incremental offset input method (U_, W_) are supported.

12. Screen Composition

? Coordinate System Setting Screen.

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel		0							002373 MM

Work Coordinate (0/1)			1: Cancel	Coord. Shift	
	G54	G55	G56	0/1	0: No Use
X	0.000	0.000	0.000	X	0.000
Y	0.000	0.000	0.000	Y	0.000
Z	0.000	0.000	0.000	Z	0.000

	G57	G58	G59	Set	Relative
X	0.000	0.000	0.000	X	0.000
Y	0.000	0.000	0.000	Y	0.000
Z	0.000	0.000	0.000	Z	0.000

Machine	
X	-15.872
Y	4.049
Z	-3.291

Relative	
X	-15.872
Y	4.049
Z	-3.291

Work Coordinate	Tool Offset					Macro Variables
-----------------	-------------	--	--	--	--	-----------------

- While checking the machine position information and relative position, you can set the target object coordinate system offset for G54~ G5, coordinate system shift, and relative coordinates.
- For more information on target object coordinate system offset setting, refer to 'Chapter 13. Setting'.

? Offset Setting Screen.



- Up to 120 sets of tool diameter (or radius) compensation and length compensation can be entered.
- In the left screen, you can check the relative position values and set the relative coordinates at an arbitrary position.

12. Screen Composition

? Macro Variable Screen. (F2→F8)

Main PG	Wheel	JOG	Tool						2004-10-21
File	Wheel		0	MLK	DRN	OPS	BDT	SBK	002373

Macro Variable(Non-Save)

Macro Variable(Save)

# 100	0.000 # 120	0.000 # 140	0.000 # 160	0.000 # 180	0.000
# 101	0.000 # 121	0.000 # 141	0.000 # 161	0.000 # 181	0.000
# 102	0.000 # 122	0.000 # 142	0.000 # 162	0.000 # 182	0.000
# 103	0.000 # 123	0.000 # 143	0.000 # 163	0.000 # 183	0.000
# 104	0.000 # 124	0.000 # 144	0.000 # 164	0.000 # 184	0.000
# 105	0.000 # 125	0.000 # 145	0.000 # 165	0.000 # 185	0.000
# 106	0.000 # 126	0.000 # 146	0.000 # 166	0.000 # 186	0.000
# 107	0.000 # 127	0.000 # 147	0.000 # 167	0.000 # 187	0.000
# 108	0.000 # 128	0.000 # 148	0.000 # 168	0.000 # 188	0.000
# 109	0.000 # 129	0.000 # 149	0.000 # 169	0.000 # 189	0.000
# 110	0.000 # 130	0.000 # 150	0.000 # 170	0.000 # 190	0.000
# 111	0.000 # 131	0.000 # 151	0.000 # 171	0.000 # 191	0.000
# 112	0.000 # 132	0.000 # 152	0.000 # 172	0.000 # 192	0.000
# 113	0.000 # 133	0.000 # 153	0.000 # 173	0.000 # 193	0.000
# 114	0.000 # 134	0.000 # 154	0.000 # 174	0.000 # 194	0.000

Work Coordinate

Tool Offset

Macro Variables

- You can set the non-storage type (#100~#199) and storage type (#200~#699) macro variables.
- If the non-storage type macro variables are used in the program, 100 variables (#100 ~ #199) are provided as the global variable. (When reset or powered OFF, they are all cleared to 0.)
- If the storage type macro variables are used in the program, #200 ~ #699 variables are provided as the global variable. (They are conserved even when powered OFF).
- To switch between the non-storage type variable items and storage type variable items, press **F8** key again.

12.6 Edit (F3)

? NC Program Editor Screen. (F3)

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									002373 MM

Wheel.nc		Absolute		X	-15.872
				Y	4.049
				Z	-3.291
		Machine		X	-15.872
				Y	4.049
				Z	-3.291

Time	16:01:57	Oper. Time	1:40:28
FEED	0.0	OVERRIDE	50
SPINDLE	1508	OVERRIDE	50

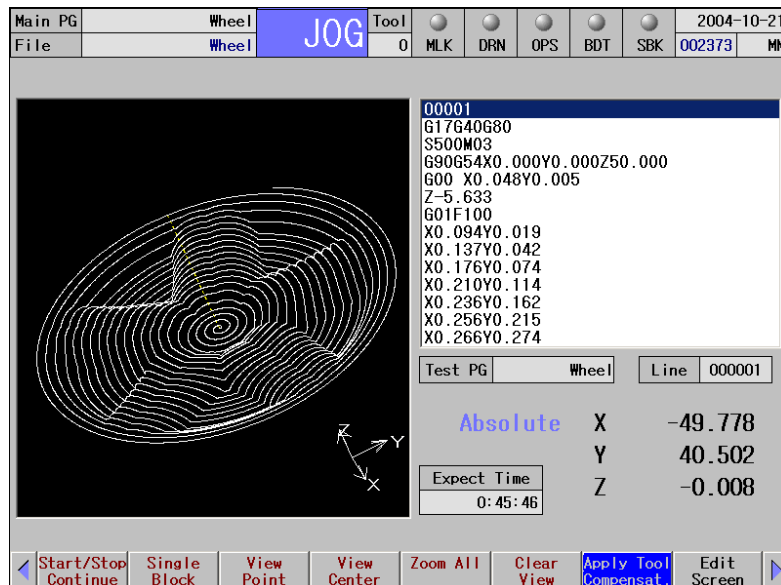
Line 000001	E:\HX3.1_MC (04.04.22)\NC\
-------------	----------------------------

Open	Save	New File	Save As	Play Back	UTIL >>	Graphic Check
------	------	----------	---------	-----------	---------	---------------

- This screen is the NC Program Editor screen.
- On the right side of the screen, you can check the current position, machine position, Feed, spindle revs and other system related information.
- Current file name is shown at the top while the current line number and the file path information are shown at the bottom.
- **F8** key is used to switch to Check Tool Path screen.
- For more information, refer to 'Chapter 6. Program Editing'.

12. Screen Composition

? Check Tool Path Screen. (F3→F8)



- Prior to the actual machining, the program machining path can be checked through the simulation.
- Using **F1**, **F2**, and **F6** key, Simulation Start, Single Block, and Simulation Cancel can be performed.
- **View Direction Setting F3** key lets you change the viewing position.
- **View Center Setting F4** key allows you to redefine the screen center.
- **Zoom All F5** key zooms in/out the tool path to fit in the screen.
- **Clear F7** key erases the tool path currently displayed in the screen.
- For more information, refer to 'Chapter 8. Graphic Function'.

12.7 Parameter (F4)

? Parameter Edit Screen. (F4)

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									002373 MM

PROGRAM	USER	MACHINING 1	MACHINING 2	SYSTEM	MACRO	AXIS	I/O SETTING	SPECIFIC	HMI
---------	------	-------------	-------------	--------	-------	------	-------------	----------	-----

NO.	Value	Unit	Comment
Macro Program			
PA 0002	1		Macro Program Edition (0:Not Allowed, 1:Allowed)
PI 0083	0		Macro program Stop on Single Block (0:Allowed, 1:Not Allowed)
PI 0084	0		Macro Program Block Display (0:Not Display, 1:Display)
PI 0105	0		Macro Program Number Called for T Code Instruction
PI 0106	0		Macro Call for T Code Instruction (0:No Use, 1:Use)
Macro Variable			
Macro Call G Code/M Code			
PI 0085	0.0		9010.nc Calling G Code
PI 0086	0.0		9011.nc Calling G Code
PI 0087	0.0		9012.nc Calling G Code
PI 0088	0.0		9013.nc Calling G Code
PI 0089	0.0		9014.nc Calling G Code
PI 0090	0.0		9015.nc Calling G Code
PI 0091	0.0		9016.nc Calling G Code
PI 0092	0.0		9017.nc Calling G Code
PI 0093	0.0		9018.nc Calling G Code
PI 0094	0.0		9019.nc Calling G Code

Move Tab Left	Move Tab Right	Move Axis Down	Move Axis Up	Find Parameter	Print
---------------	----------------	----------------	--------------	----------------	-------

- This screen is used to set or check various parameters required for the system operation.
- This screen is composed of 10 items including 'Program', 'User', 'Machining 1', 'Machining 2', 'System', 'Macro', 'Axis', 'I/O Setting', 'Special Functions', and 'HMI'.
- Each item (tab) can be selected by the mouse click or using **F1**, **F2** keyboard keys.
- 'Axis' item is composed of 32 axes. To change the axis to set from 'Axis' item(tab), use **F3**, **F4** keyboard keys. (If 'Axis' item is activated, current axis information is displayed at the top right side of the screen.)
- **F7** **Print** menu is used to make printout of the currently activated item. This menu can be used only when the password of more than 1st step is entered in the password item.
- **F8** **Servo Drive** menu will guide you to SERCOS interface type 'Servo Drive' related setting screen.
- Refer to 'Chapter 10. Parameter Edit'.

12. Screen Composition

12.8 Utility (F5)

? DNC File Transmission/Reception Screen. (F5→F1)

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									002373 MM

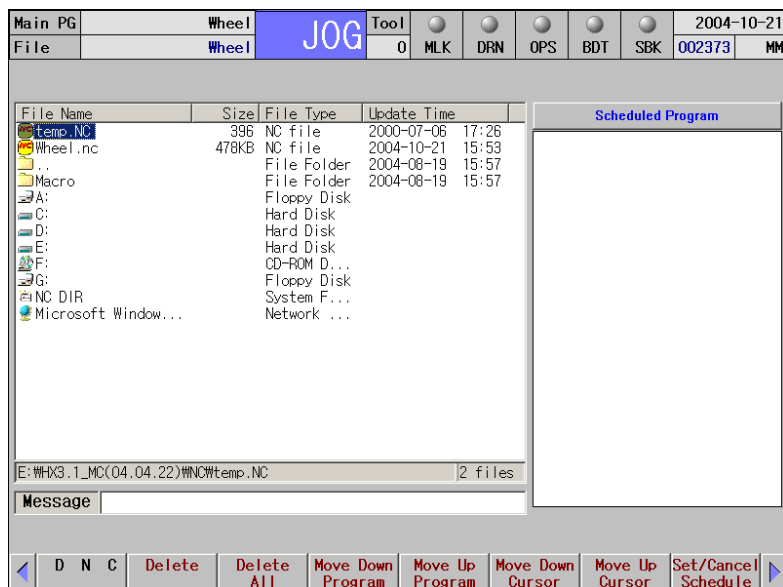
File Name		Size (Byte)	
Path			

Status		Size (Byte)	
Rate (%)			
Block			

D N C	Send Program	Receive Program	Schedule Program
-------	--------------	-----------------	------------------

- This screen is used to perform the transmission/reception of NC program file through the RS 232C communication.
- For more information, refer to 'Chapter 11. Utility'.

? Scheduling Screen. (F5→F6)



- This screen is used to set the sequence of multiple program machining for the automatic uninhabited machining.
- For more information, refer to 'Chapter 3. Automatic Operation'.

12. Screen Composition

12.9 System Management (F6)

? Alarm Screen. (F6→F1)

Main PG	Wheel	AUTO	Tool						2004-10-21
File	Wheel		0	MLK	DRN	OPS	BDT	SBK	000000
G_90000[1/ 1] : EMERGENCY STOP									
NO.	CODE	MESSAGE					DATE	TIME	
1	F_84912	#4 Axis SERVO Alarm					04/10/21	16:06:46	
2	F_84411	#2 Axis Encoder C Phase Alarm					04/10/21	16:06:49	
3	F_84512	#2 Axis SERVO Alarm					04/10/21	16:06:49	

Alarm List

Password

? Status Information Screen. (F6→F4)

Main PG	Wheel	JOG	Tool	0	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel									000000 MM

Status Info 1	Status Info 2	Status Info 3	Specific	Monitoring
---------------	---------------	---------------	----------	------------

NO.	Value	Unit	Comment
IPR Information			
SV 11	0		Operation Status (0:Stop 1:Run)
SV 12	693		IPR Heart Bit
SV 15	0		Wait EOM (0:No wait 1:Wait)
SV 16	0		AF Usage
S/W Module Information			
SV 20	0		IPO RUN status
SV 21	28		IPO Heart Bit
SV 22	48507		POS Heart Bit
SV 23	24253		PLC Heart Bit
SV 30	194032		Timer
Version Information			
SV 3	0.16		MMI Version
SV 4	0.00		MMI Debug Version
SV 13	3.00		IPR Version
SV 17	3.00		IPR Debug Version
SV 24	2.400		IPO Version
SV 25	2.400		POS Version

Move Tab Left	Move Tab Right	File Information					System Manager
---------------	----------------	------------------	--	--	--	--	----------------

- This screen displays the system status monitoring information.
- This screen can be viewed only when the password of more than 1st step is entered in the Password screen.
- This screen includes 4 items such as 'Status Info. 1', 'Status Info. 2', 'Special Functions', and 'Monitoring' and each item is further itemized.
- Each item is set to Read-Only state that cannot be modified and therefore, only the monitoring is possible.
- Switching between items can be done by the mouse click or by using **F1**, **F2** keys.

12. Screen Composition

? Servo Wave Screen. (F6→F5)

Various machine related information such as axis feed, machine position of each axis, and contact status within the system, can be monitored through the graphic screen in real time. Through this graphic presentation, servo adjustment and self-diagnosis can be made.



- This screen can be viewed only when the password of more than 1st step is entered in the Password screen.
- Through Servo Graph Mode and PLC Timing Chart, required information can be effectively diagnosed.
- Using the Trigger function and Rising Time function, required information can be effectively diagnosed.
- **F1** key can be used to start or stop the servo wave.
- **F2** key can be used to save the graph contents to a file. File is stored under the HX folder.
- **F3** key can be used to automatically adjust the screen scale.
- **F4 / F5** key and cursor key combination can be used to check the value at a certain point or enlarge a certain portion of graph.
- **F6** key can be used to adjust and determine the screen mode, graph type, usage status, and scale.
- **F7** key is used to select the Trigger function and Rising Time function.
- Using the cursor key, the scale and screen can be adjusted to fit your need.

? Diagnosis Screen. (F6→F6)

Main PG	Wheel	JOG	Tool	MLK	DRN	OPS	BDT	SBK	2004-10-21
File	Wheel	0	MLK	DRN	OPS	BDT	SBK	000577	MM

X-Y	G-F	R	T	C	D	USER
-----	-----	---	---	---	---	------

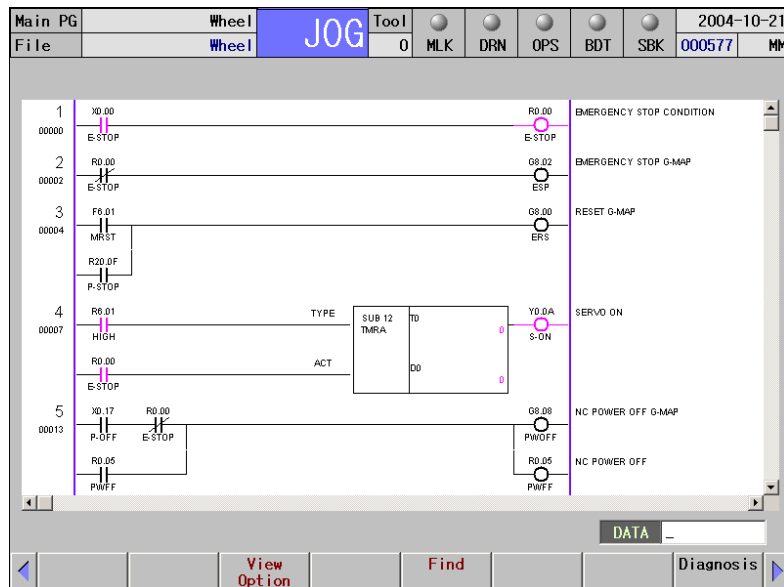
NO.	0x18-	0x10-	0x08-	0x00	NO.	0x18-	0x10-	0x08-	0x00
X 0000	00010111-01000000-00000000-01000101				Y 0000	00000000-00000000-00010100-00000000			
X 0001	00000000-00000000-00000000-00000000				Y 0001	00000000-00000000-00000000-00000000			
X 0002	00000000-00000000-00000000-00000000				Y 0002	00000000-00000000-00000000-00000000			
X 0003	00000000-00000000-00000000-00000000				Y 0003	00000000-00000000-00000000-00000000			
X 0004	00000000-00000000-00000000-00000000				Y 0004	00000000-00000000-00000000-00000000			
X 0005	00000000-00000000-00000000-00000000				Y 0005	00000000-00000000-00000000-00000000			
X 0006	00000000-00000000-00000000-00000000				Y 0006	00000000-00000000-00000000-00000000			
X 0007	00000000-00000000-00000000-00000000				Y 0007	00000000-00000000-00000000-00000000			
X 0008	00000000-00000000-00000000-00000000				Y 0008	00000000-00000000-00000000-00000000			
X 0009	00000000-00000000-00000000-00000000				Y 0009	00000000-00000000-00000000-00000000			
X 0010	00000000-00000000-00000000-00000000				Y 0010	00000000-00000000-00000000-00000000			
X 0011	00000000-00000000-00000000-00000000				Y 0011	00000000-00000000-00000000-00000000			
X 0012	00000000-00000000-00000000-00000000				Y 0012	00000000-00000000-00000000-00000000			
X 0013	00000000-00000000-00000000-00000000				Y 0013	00000000-00000000-00000000-00000000			
X 0014	00000000-00000000-00000000-00000000				Y 0014	00000000-00000000-00000000-00000000			
X 0015	00000000-00000000-00000000-00000000				Y 0015	00000000-00000000-00000000-00000000			
X 0016	00000000-00000000-00000000-00000000				Y 0016	00000000-00000000-00000000-00000000			
X 0017	00000000-00000000-00000000-00000000				Y 0017	00000000-00000000-00000000-00000000			

Move Tab Left	Move Tab Right	Add/Delete User Item	Sequence View	System Manager
---------------	----------------	----------------------	---------------	----------------

- Various system related contact information can be checked.
- This screen is composed of 6 items of 'X-Y', 'G-F', 'R', 'T', 'C', and 'D' as well as 'USER' item for separate monitoring of selected items.
- Each item can be selected by the mouse click or using **F1 / F2** key of keyboard or MDI.
- USER Item Addition
 - ① On top of the item to monitor, press **F4** key that is the Add/Remove USER Item menu.
 - ② Repeat the above step ① on the other item to monitor.
 - ③ Move to 'USER' item(Tab) to see the added contents.
- To delete an User item, move to 'USER' item and press **F4** key on an item to delete.
- The contact contents displayed on the screen can be added or removed through simple external file manipulation.

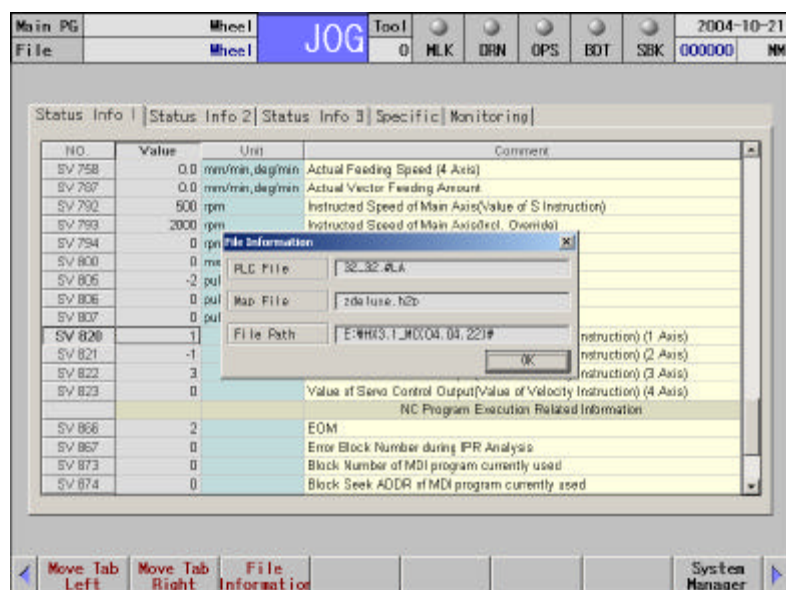
12. Screen Composition

? Sequence (Ladder) Screen. (F6→F6→F7)



- Information on each contact can be graphically checked.
- Various contact display of the currently running system and the ladder operation sequence can be checked in real time.
- To search a contact, enter the keyword in the order to 'Address Name – Number - Bit'. For example, to view the 11th contact of X100, enter 'X100.0A'. Bit must be entered in 2 hexadecimal digits.
- To continuously search the same address, press ENTER key.

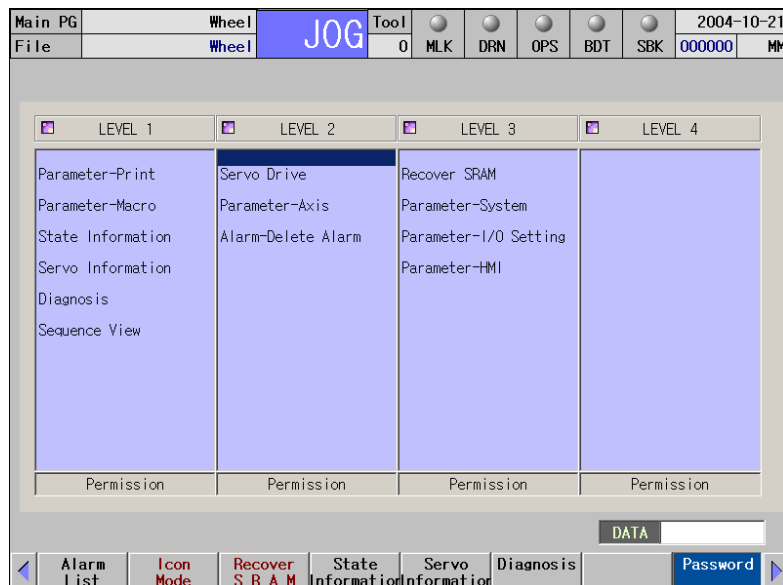
? Version Screen. (F6→F7)



- In this screen, you can check the version information on all modules comprising the HX system.

12. Screen Composition

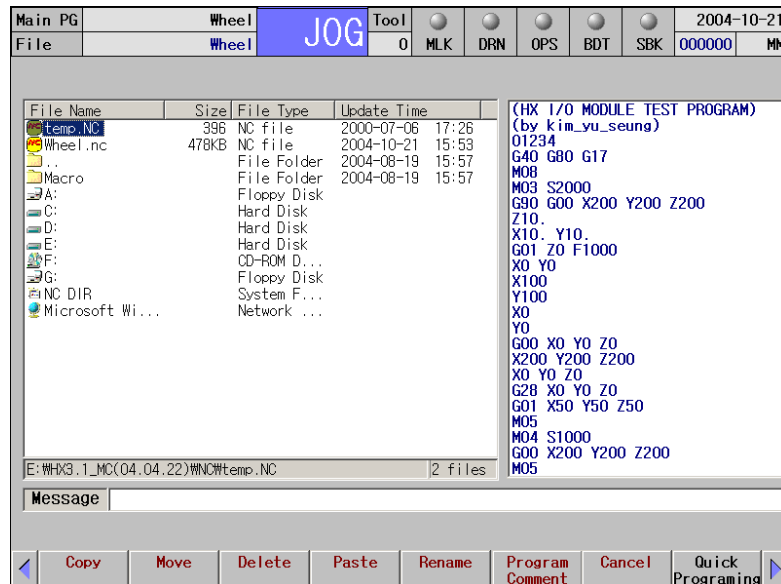
? Password Input Screen. (F6→F8)



- Depending on the usage, some functions of the system are protected by the password. To use such functions, you must enter the corresponding password.
- Passwords cannot be altered and refer to 'Chapter 9. System Management' for the passwords.
- When you select this menu, Hide Screen menu that hides the HX screen to an icon is displayed. (F2)

12.10 Program Management (F8)

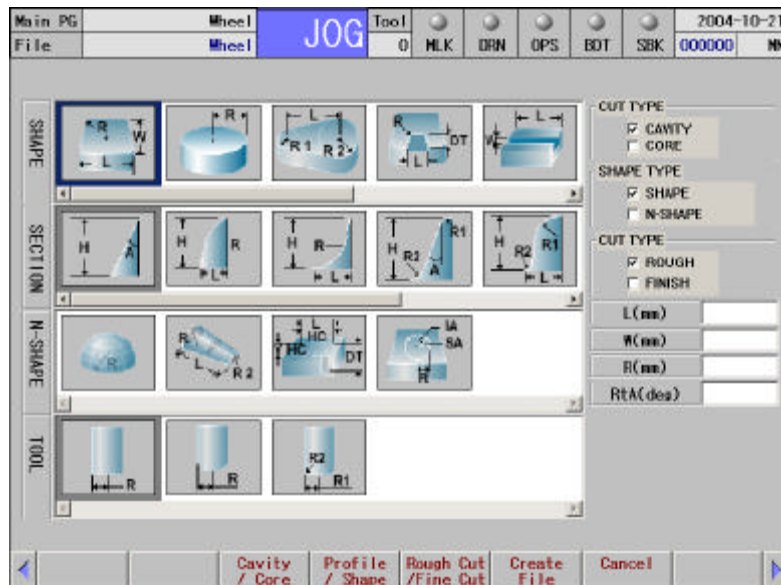
? Program Management Screen. (F8)



- This screen is used to select the machining program.
- Besides the program selection, you can use other shell functions (Copy, Move, and Rename).
- You can add, change or check the description for each program.
- Front part of the selected program is shown on the right pane and the simple comment written by the user is also provided at the bottom message section.
- For more information, refer to 'Chapter 7. Program Management'.

12. Screen Composition

? Interactive Program Screen. (F8→F8)



- The program for basic Cavity and Core shapes can be easily and interactively created.
- The programs created in this screen are stored in 'COCA' folder under NC folder of HX System.
- For more information, refer to 'Chapter 13. Interactive Program'.

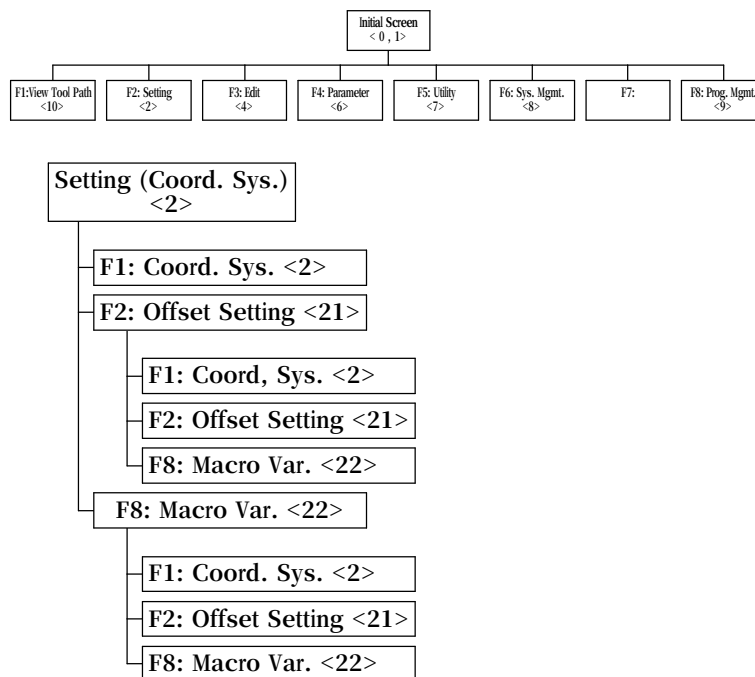
12.11 Screen HELP Function (SHIFT+F12)

? Screen HELP Display

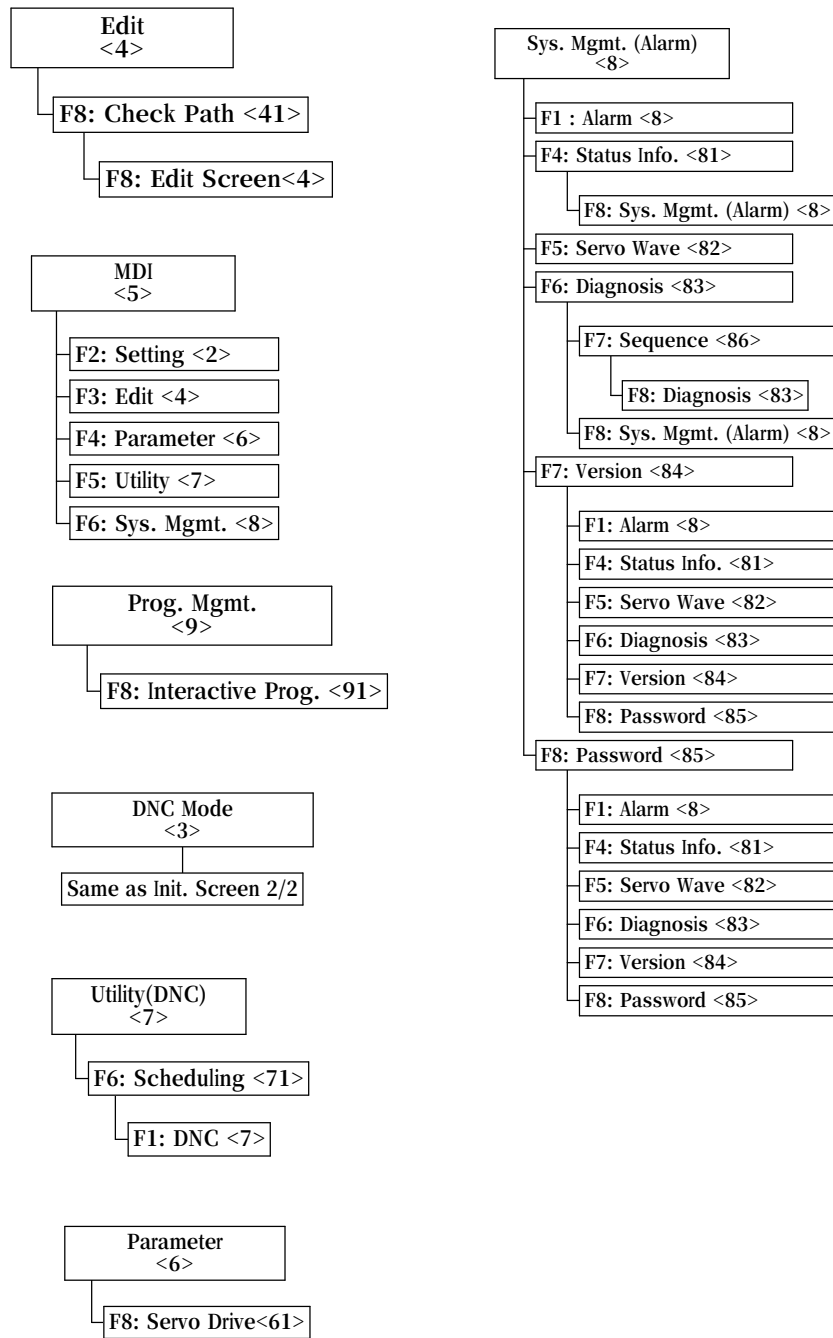
In HX system, each screen has a unique screen number. If you press 'SHIFT + HELP' key on a particular screen, corresponding HELP is provided.

- The HELP file corresponding to each screen is stored in a 'SN(Screen Number).htm' format in '/Help/Screen' folder under the system folder. You can add new HELP files easily if you compose and use new screens.
- The default screen numbers are as shown below and the user can add new screens. (Refer to Basic MMI section in Appendix of Maintenance Manual for more information.)

HX 2.1 Screen Composition



12. Screen Composition





Memo



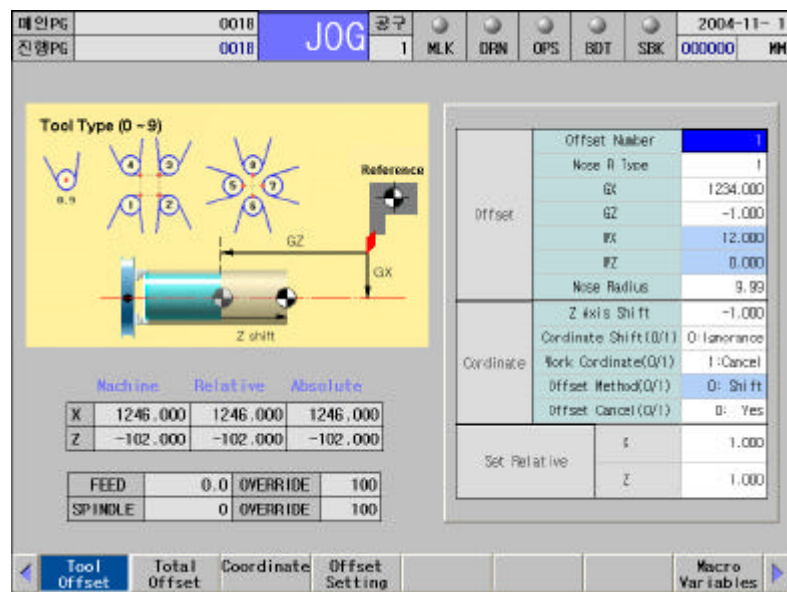
13. Setting

- 13.1 Tool offset(Lathe)
- 13.2 Total offset(Lathe)
- 13.3 Coordinate
- 13.4 Offset setting
- 13.5 Macro variable

13.1 Tool Offset (Lathe)

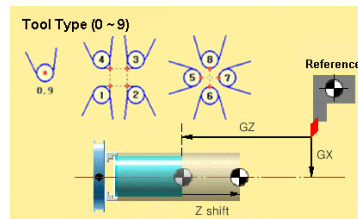
This screen is to set tool offset for each offset number of total 64 sets. On this screen, the reference position of the work-piece coordinate system and the information on tool thread end are set.

It is possible to set whether to use coordinate shift or not and the relative coordinates at an arbitrary location.



■ How to operate

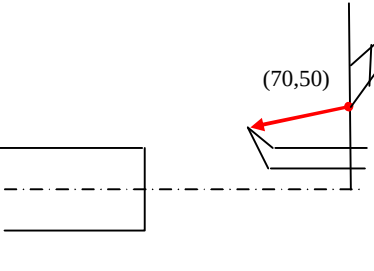
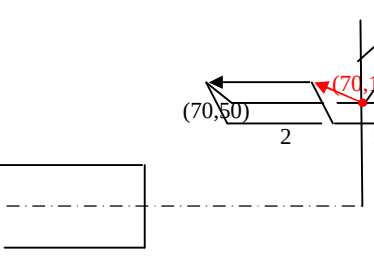
1. Press the menu of **Set**, F2 in the Initial Menu.
2. Seeing the figure in the below, input the followings. For input, use a direction key to select an item, input the value and then, press ENTER.
3. Offset number: Input an offset number to correct out of 1 – 64. In case it is 0, other offset values can't be input.



4. Shape of tool thread end: Referring to the number of the thread end shape in the above figure (0-9), input the relevant thread end number. In general, the tool for rough/minute machining of external diameter is 3 and the tool for boring of internal diameter is 2.
5. GX: As shown on the above figure, input the distance of X on the machine coordinate system to the reference position of X on the work-piece coordinate system. It is good to directly input the current machine location as default. In case of input with incremental value, if you input, together with I just like I-30, it is input like [Machine location + Input value (-30)]. The offset amount for X wear for the selected offset number becomes cleared to be 0.
6. GZ: As shown on the above figure, input the distance of Z on the machine coordinate system to the reference position of the work-piece coordinate system. The method of input is same as for GX. The offset amount for Z wear for the selected offset number becomes cleared to be 0.
7. Radius of tool thread end: Input the radius of the tool thread end.
8. Shift of Z: As shown on the above figure, input as much as movement of Z reference position of the work-piece coordinate system. The item of Z offset helps you move the reference position of the work-piece as much as the input value without resetting the coordinate system when you use other work-piece whose length is different from the initial material after setting the work-piece coordinate system. In case the latter material is shorter than that first set with the coordinate system, the value becomes - . This value is applied to the offset value of Z only in

case of using coordinate system shift.

9. How to apply offset: This is the parameter for applying offset. In case it is 0, offset application becomes in the type of shift, and in case it is 1, it becomes in the type of axis feed.

Tool offset is performed as the shift function	 (70,50) (70,100)→(70,90) After working on N5 block, only the current coordinate value is changed to (70, 90) and there is no axis feed.
Tool offset is performed as feed action	 1 (70,50) (70,100) 2 In case of working on N5 block, X feeds by 5mm and Z, by -10mm. At this time, the current location (70, 100) is maintained as it is.

How to set automatic tool offset

1. Move the cursor to the item of Offset Number on the screen and select the tool offset number.
2. After slightly machining the section of a work-piece by use of the relevant tool, stop the operation without changing the location of Z (machine location).
3. Locate the cursor on GZ item of tool offset and input [I0] [INPUT].
4. Do slight machining for the external diameter of the work-piece by use of the same tool, and feed the tool to Z without changing the location of X (machine location).
5. Measure the diameter of the work-piece.
6. Locate the cursor on GX of tool offset.
7. In consideration of the measured diameter of the work-piece and of the

machining direction of the tool, input [I] [-Diameter] [INPUT] for GX. At this time, diameter should be input with incremental value. And then, the tool offset for X is automatically set as shown on the below example.

Ex)

13.2 Total Tool Offset (Lathe)

This screen is to set tool offset and wear compensation data for each offset number of total 64 sets. On this screen, it is possible to change the tool offset data, compensation data of tool wear, compensation data of tool thread end, etc.

On this screen, absolute and incremental input are possible. Absolute input is the function to directly change the offset value on the location where the cursor is, and incremental input (command with U__, W__) should be done by adding/subtracting the input value to/from the offset data of the location where the cursor is. Incremental input is usually used for input of wear compensation data.

메인PG	001B	JOG	공구	1	MLK	OPN	OPS	BDT	SBK	2004-11- 1
진행PG	001B									000000

Tool Type (0 ~ 9)

Reference

Z shift

	Machine	Relative	Absolute
X	1246.000	1246.000	1246.000
Z	-102.000	-102.000	-102.000

FEED	0.0	OVERRIDE	100
SPINDLE	0	OVERRIDE	100

Offset	Offset Number	1
	Nose R Type	1
	GX	1234.000
	GZ	-1.000
	WX	12.000
Coordinate	WZ	0.000
	Nose Radius	9.99
	Z Axis Shift	-1.000
	Coordinate Shift (Q1)	0: Ignoreance
	Work Coordinate (Q1)	1: Cancel
Set Relative	Offset Method (Q1)	0: Shift
	Offset Cancel (Q1)	0: Yes

Tool Offset	Total Offset	Coordinate	Offset Setting				Macro Variables
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13.3 Coordinate System

This is the screen to set offset values of coordinate system from G54 to G59. On the work-piece coordinate system, the offset data of the machine coordinate against the reference position of the work-piece coordinate system is set. In order to normally use the function of selecting work-piece coordinate system (G54~G59), you should input this offset data before machining.

The function - work-piece coordinate system 0: maintain/1: cancel – is the parameter to set the function of maintaining the offset value of the work-piece coordinate system which is set at the time of RESET or of canceling. If it is set to maintain the work-piece coordinate system, the last offset value remains even when the initial power is on.

It is possible to set the amount of coordinate system shift and relative coordinate.

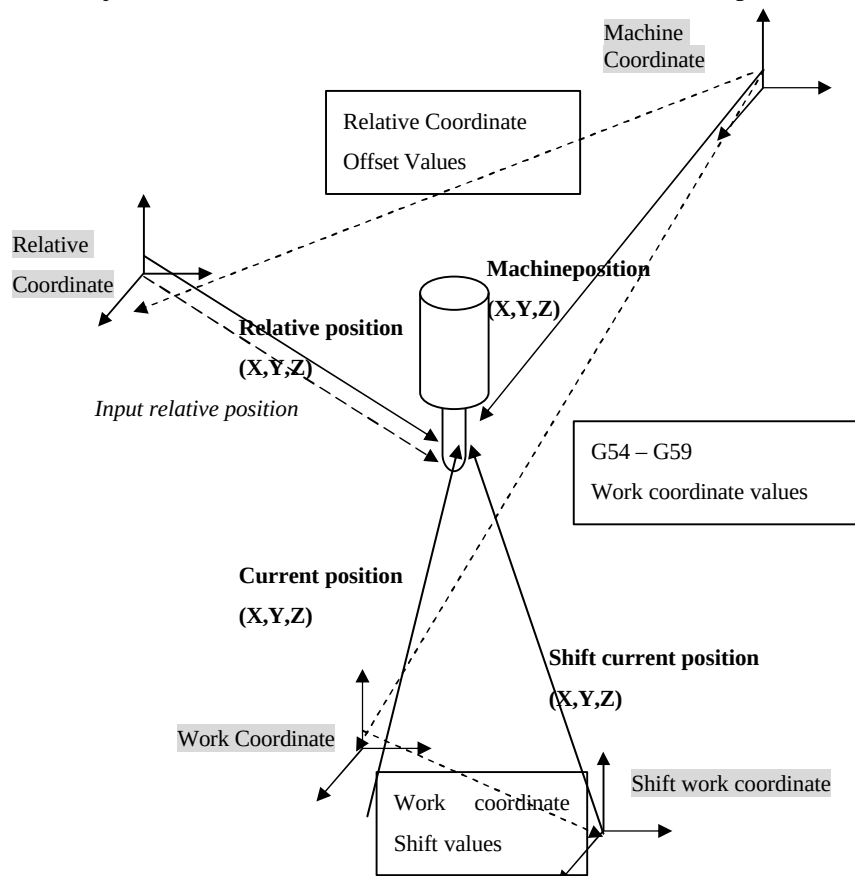
How to operate

1. Press the menu of **Set**, **F2** on Initial Screen.

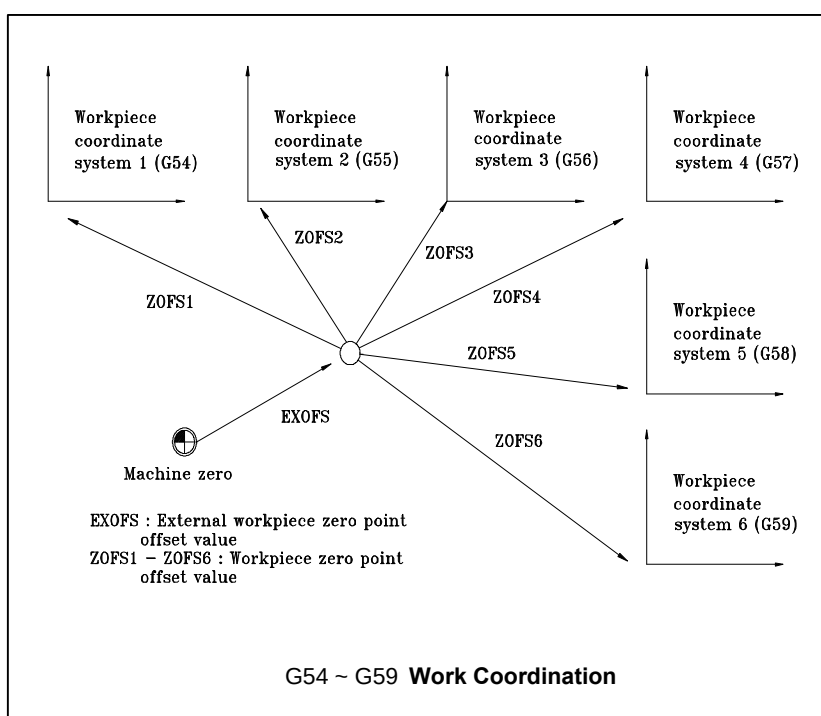
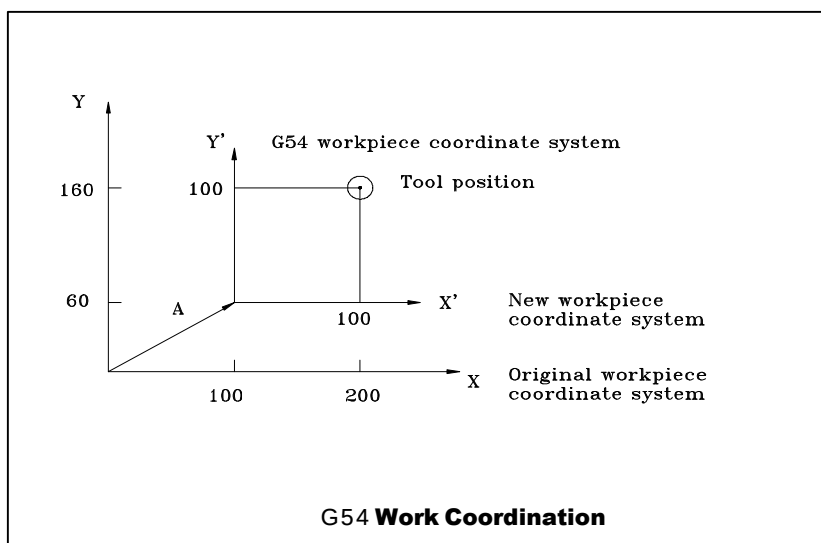
Work Coordinate (0/1)			1: Cancel	Coord. Shift	
	G54	G55	G56	0/1	0: No Use
X	0.000	0.000	0.000	X	0.000
Y	0.000	0.000	0.000	Y	0.000
Z	0.000	0.000	0.000	Z	0.000
	G57	G58	G59	Set Relative	
X	0.000	0.000	0.000	X	0.000
Y	0.000	0.000	0.000	Y	0.000
Z	0.000	0.000	0.000	Z	0.000

2. At the right, the coordinates are indicated and at the left, the above input window appears.

3. In order to input coordinate values of G54~G59, locate the blue cursor on the position you want, input the offset value of the machine location and then, press **Enter**.
4. In order to input incremental value for the machine location at the time of input of G54~G59, input I and then, a number next to I. That is, if you input 'I10', the value of 10 added to the current machine location becomes the offset value of the coordinate system. This function is useful in case of inputting the location value increased by the tool's radius as offset after touching the side of the tool on the work-piece.
5. In case of setting the coordinate shift and relative coordinate, move the cursor to the location that you want and then, set with an arbitrary value.
6. The function of setting the relative coordinate is to set the current machine location with arbitrary coordinate system. It is used to check the relative distance between the standard tool and other tool in manual mode and then, to find offset value. For example, if you want to set the location where the tool is now at (0, 0, 0) as relative location, input 0 → Enter for each axis in the item of Relative Coordinate Setting.



7. The coordinate shift value is not applied in case of 'Ignore Shift'. Only in case of 'Use Shift', the shift offset is applied when giving a command to select work-piece coordinate system (G54~G58) or to set work-piece coordinate system. In order to change the coordinate shift use (1) and the coordinate shift ignore (0), move the cursor, input 0 or 1 and then, enter.





13.4 Set Offset

This screen is to input the compensation value of tool diameter and of tool length of total 128 sets.

The input compensation value of diameter is used for the command of Tool Diameter Compensation using G41, G42 on the program. For designating the compensation value, use Address D. According to Parameter PI[72] 'How to apply compensation value of tool diameter (0: apply with diameter, 1: apply with radius)', you can selectively use input of diameter value or input of radius value.

The offset data of tool length compensation is the length compensation data that is applied for the command of tool length compensation (G43~G44). Based on Address H, you can select offset number. On the left of this screen, it is possible to set relative coordinate.

How to operate

- ① Press the menu of **Set**, **F2** on the Initial Screen and the menu of **Set Offset**, **F2**.

TOOL DIA.		TOOL HEIGHT		TOOL DIA.		TOOL HEIGHT	
D 1	0.000	H 1	5.000	D 65	0.000	H 65	0.000
D 2	0.000	H 2	5.000	D 66	0.000	H 66	0.000
D 3	0.000	H 3	0.000	D 67	0.000	H 67	0.000
D 4	0.000	H 4	0.000	D 68	0.000	H 68	0.000
D 5	0.000	H 5	0.000	D 69	0.000	H 69	0.000
D 6	0.000	H 6	0.000	D 70	0.000	H 70	0.000
D 7	0.000	H 7	0.000	D 71	0.000	H 71	0.000
D 8	0.000	H 8	0.000	D 72	0.000	H 72	0.000
D 9	0.000	H 9	0.000	D 73	0.000	H 73	0.000
D 10	0.000	H 10	0.000	D 74	0.000	H 74	0.000
D 11	0.000	H 11	0.000	D 75	0.000	H 75	0.000
D 12	0.000	H 12	0.000	D 76	0.000	H 76	0.000
D 13	0.000	H 13	0.000	D 77	0.000	H 77	0.000
D 14	0.000	H 14	0.000	D 78	0.000	H 78	0.000
D 15	0.000	H 15	0.000	D 79	0.000	H 79	0.000

- ② Move the cursor to the tool number to input, input the value and press **Enter** , and the input value will be set.

13.5 Macro Variable

This is to set macro variable used in the NC program. Macro variable is divided into two types: for no-saving and for saving. Macro variable for saving means the variable that previous value is saved even after the system is closed and the macro variable for no-saving becomes cleared when the program is closed or RESET.

How to operate

- ① Press the menu of **Set** , **F2** and the menu of **Set Offset** , **F8** .

Macro Variable(Non-Save)			Macro Variable(Save)		
# 100	0.000	# 120	0.000	# 140	0.000
# 101	0.000	# 121	0.000	# 141	0.000
# 102	0.000	# 122	0.000	# 142	0.000
# 103	0.000	# 123	0.000	# 143	0.000
# 104	0.000	# 124	0.000	# 144	0.000
# 105	0.000	# 125	0.000	# 145	0.000
# 106	0.000	# 126	0.000	# 146	0.000
# 107	0.000	# 127	0.000	# 147	0.000
# 108	0.000	# 128	0.000	# 148	0.000
# 109	0.000	# 129	0.000	# 149	0.000
# 110	0.000	# 130	0.000	# 150	0.000
# 111	0.000	# 131	0.000	# 151	0.000
# 112	0.000	# 132	0.000	# 152	0.000
# 113	0.000	# 133	0.000	# 153	0.000
# 114	0.000	# 134	0.000	# 154	0.000

- ② It is possible to directly move between the macro variable for saving and that for no-saving by use of mouse, and also possible to move them by pressing **F8** .
- ③ After moving the cursor to where to set, input the value and press **Enter** , and the input value is set.
- ④ 100 macro values for no-saving from #100 to #199 are supported, and 500 macro variables for saving from #200 to #699 are supported.



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